

NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

**Research Project Report on the Topic:
Network Topology and Systemic Risk
in Financial Markets**

Fulfilled by:

Student of the Group БПАД241

Chistiakova Maria Ruslanovna

31.05.2026

(date)

Assessed by the Project Supervisor:

Lukianchenko Petr Pavlovich

Senior Lecturer

Faculty of Computer Science, HSE University

Contents

1	Abstract	2
2	Keywords	3
3	Introduction	4
4	Overview of Bibliographical Sources	4
5	Methodology: Model Specification	5
5.1	Baseline order-driven market	5
5.2	One-dimensional interaction structure	6
5.3	Stress, effective thresholds, and failure	6
5.4	Shock mechanisms	7
5.5	Outcome metrics	8
6	Simulation Design and Dataset	9
7	Empirical Results	10
7.1	Repeated market shocks versus the single benchmark	10
7.2	Spatial diagnostics under cascade and local cash shocks	15
7.3	Sensitivity to interaction radius, contagion, and thresholds	19
7.4	Frequency scan for repeated relative shocks	24
8	Statistical Validation and Hypothesis Evaluation	31
8.1	Validation of H1: localized shocks and propagation beyond the initial region . . .	31
8.2	Validation of H2: repeated shocks and benchmark-level damage	32
8.3	Validation of H3a–H3c: thresholds, contagion strength, and interaction radius . .	33
8.4	Final statistical decision	34
9	Critical Evaluation and Further Research	35
10	Conclusion	36
11	References	38
	Appendix A: Disclosure of AI assistance	40

1 Abstract

This report studies cascading shocks in a one-dimensional agent-based financial market with an explicit order book, heterogeneous traders, threshold-based failure, and forced liquidation. The central research problem is whether localized or temporally clustered disturbances can generate systemic damage when each individual shock is relatively small.

The study makes three contributions. First, the baseline simulator is extended with repeated price, liquidity, and local cash-shock events, allowing direct comparison between one benchmark shock and a sequence of smaller shocks with comparable nominal magnitude. Second, a cascade layer is added: each trader receives a spatial position, a capital-drawdown stress variable, an individual resistance threshold, and an effective threshold affected by neighboring failures. Third, the empirical analysis combines visual diagnostics, seed-level comparisons, controlled parameter sweeps, and statistical validation of the main hypotheses.

The final validation gives a nuanced result. The localized-propagation hypothesis is not supported by the controlled ablation test: the propagation-enabled local-shock baseline does not generate larger outside-region damage than the propagation-suppressed control. The repeated-shock hypothesis is partially supported: repeated small relative shocks reproduce benchmark-level damage in several regimes under a pre-specified tolerance rule, but this is interpreted as benchmark-like comparability rather than formal equivalence. The threshold-resilience component is statistically supported: higher mean resistance thresholds generally reduce cascade probability, cascade size, outside-region failures, and spatial extent, although the relationship is not strictly monotone at every threshold level. Overall, the results indicate that systemic fragility in the implemented model is driven more robustly by shock timing and agent-level resistance than by the tested local contagion channel.

В отчёте исследуются каскадные шоки в одномерной агентной модели финансового рынка с явной книгой лимитных заявок, разнородными трейдерами, пороговым механизмом отказа и вынужденной ликвидацией. Центральным исследовательский вопрос состоит в том, способны ли локализованные или сгруппированные во времени возмущения приводить к системному ущербу, если каждое отдельное воздействие относительно мало.

В работе представлены три основных результата. Во-первых, базовый симулятор расширен повторяющимися ценовыми, ликвидностными и локальными денежными шоками, что позволяет сравнивать одиночный benchmark-шок с последовательностью меньших воздействий сопоставимой номинальной величины. Во-вторых, добавлен каскадный слой: каждому агенту задаются пространственная позиция, стресс как относительная просадка капитала, индивидуальный порог устойчивости и эффективный порог, зависящий от отказов соседей. В-третьих, эмпирическая часть объединяет визуальную диагностику, сравнения по случайным seed, контролируемые переборы параметров и статистическую проверку гипотез.

Итоговая валидация даёт неоднозначный, но содержательный результат. Гипотеза о локальном распространении не подтверждается контролируемым абляционным тестом: базовый сценарий с включённым распространением не создаёт больший ущерб вне исходной

области шока, чем контроль с подавленным распространением. Гипотеза о повторяющихся шоках подтверждается частично: малые повторяющиеся относительные шоки в ряде режимов воспроизводят уровень ущерба benchmark-сценария в пределах заранее заданного допуска. Компонент гипотезы, связанный с порогами устойчивости, статистически подтверждён: более высокие средние пороги в целом снижают вероятность каскада, его размер, отказы вне исходной области и пространственный охват. В целом результаты показывают, что в реализованной модели системная хрупкость устойчивее связана с временной структурой шоков и порогами агентов, чем с протестированным локальным каналом заражения.

2 Keywords

agent-based modeling; cascading shocks; one-dimensional lattice; limit order book; threshold failure; forced liquidation; repeated shocks; systemic risk; fixed-volume spread; statistical validation.

3 Introduction

Financial markets are interaction systems in which local disturbances can be amplified by trading behavior, liquidity conditions, and balance-sheet constraints. A shock that is small at the level of an individual agent may become systemic if it forces liquidations, changes prices, and increases the stress of other agents. For this reason, market fragility is not determined only by the magnitude of a single shock. It also depends on how shocks are distributed in time, how agents are connected, and how much capital loss each agent can absorb before leaving the market.

This report deliberately studies a one-dimensional interaction structure. The purpose is not to reproduce the full topology of real financial systems, but to obtain a controlled environment in which the origin, direction, and extent of propagation are observable. In a one-dimensional lattice, the model can distinguish local damage from a genuine cascade that travels beyond the initially affected interval.

The objective of the project is to evaluate how shock timing, agent-level resistance thresholds, and local interaction rules affect cascade formation in an order-driven agent-based market. The study is organized around three research questions:

- RQ1. Can localized or repeated shocks trigger failures outside the initially affected set of agents?
- RQ2. Does the temporal frequency of repeated shocks change the severity of damage beyond total shock magnitude alone?
- RQ3. How do resistance thresholds, contagion strength, and interaction radius affect final cascade size and cascade probability?

The corresponding hypotheses are:

- H1. Spatially localized micro-shocks can initiate cascade failures beyond the initially perturbed interval through endogenous liquidation and local contagion.
- H2. Repeated small shocks can reproduce or exceed the damage of a single benchmark shock when the inter-shock interval is short enough.
- H3. The resilience mechanism is evaluated in three parts: H3a, higher resistance thresholds reduce cascade severity; H3b, stronger contagion increases fragility; and H3c, wider interaction neighborhoods increase fragility.

4 Overview of Bibliographical Sources

The literature provides several complementary explanations of cascade formation. Threshold models of collective behavior show that heterogeneous individual thresholds can generate abrupt aggregate transitions even when the initial disturbance is limited [1, 5]. Informational-cascade models explain how agents may ignore private information and follow observed actions,

producing inefficient collective outcomes [2, 10]. These models are important because they formalize the logic of cascade amplification, but they usually abstract from market microstructure, forced liquidation, and order-book liquidity.

A second line of research uses agent-based financial markets to explain endogenous volatility and non-equilibrium price dynamics. Models by Lux and Marchesi, Cont and Bouchaud, Farmer and Joshi, and Chiarella et al. demonstrate that heterogeneous trading strategies can generate fat-tailed returns, volatility clustering, and large price fluctuations without assuming large external shocks [3, 4, 6, 7]. These studies motivate the order-driven agent-based approach used in this report. However, their main focus is usually on endogenous market fluctuations or herding, whereas the present work focuses on controlled shock timing, local propagation, and threshold-based failure.

A third related stream studies systemic risk in financial networks. Interbank and asset-network models show that local shocks can propagate through balance-sheet exposures, overlapping portfolios, and endogenous deleveraging [8, 9, 13, 14, 16]. These models are valuable for understanding high-dimensional contagion, but their complexity can make it difficult to isolate the basic spatial mechanism by which a localized disturbance becomes non-local.

The research gap addressed here is the joint analysis of repeated shock timing, threshold contagion, and forced liquidation inside an order-driven agent-based market. Existing threshold-cascade papers explain why local failures may spread, but they do not model limit-order-book liquidation feedback. Existing financial ABMs reproduce volatility and fat tails, but they usually do not isolate repeated-shock frequency against a single benchmark shock. Network-contagion papers capture high-dimensional exposures, but they do not provide a minimal spatial laboratory in which the propagation path is directly visible.

In contrast to threshold models, the model contains a limit order book and liquidation feedback. In contrast to many financial ABMs, it explicitly compares repeated shocks with a single benchmark shock. In contrast to high-dimensional network-contagion models, it uses a one-dimensional lattice to make spatial propagation transparent. The contribution is therefore methodological: the model provides a minimal but interpretable framework for studying how repeated local disturbances become systemic.

5 Methodology: Model Specification

5.1 Baseline order-driven market

The model is built on an order-driven market with a limit order book. The exchange stores bid and ask orders, matches compatible orders, computes the mid-price, and generates a stochastic dividend process. The market is populated by heterogeneous traders. Random traders generate background order flow. Fundamentalists compare the current price with a dividend-based valuation and tend to stabilize the price. Chartists react to price dynamics and sentiment and may amplify trends. Universalists switch between fundamentalist and chartist behavior. Market makers supply bid and ask quotes and affect liquidity through the depth and tightness of the book.

Let P_t denote the mid-price at iteration t , defined through the best bid B_t and the best ask A_t :

$$P_t = \frac{A_t + B_t}{2}. \quad (5.1)$$

The portfolio value of trader i is

$$E_i(t) = c_i(t) + q_i(t)P_t, \quad (5.2)$$

where $c_i(t)$ is cash and $q_i(t)$ is the asset position. This value is used to define stress and failure in the cascade layer.

5.2 One-dimensional interaction structure

Traders are placed on a one-dimensional lattice indexed by $i = 1, \dots, N$. For a fixed interaction radius R , the neighborhood of agent i is

$$\mathcal{N}_i(R) = \{j : 0 < |j - i| \leq R\}. \quad (5.3)$$

The lattice can be interpreted as a reduced-form representation of local exposure, local information, or behavioral proximity. A one-dimensional structure is deliberately restrictive, but it makes the propagation path observable.

Each agent has a binary failure state

$$F_i(t) \in \{0, 1\}. \quad (5.4)$$

The failure time is

$$t_{\text{fail},i} = \min\{t : F_i(t) = 1\}, \quad (5.5)$$

with $t_{\text{fail},i} = +\infty$ if the agent never fails.

5.3 Stress, effective thresholds, and failure

The model measures agent stress as relative capital drawdown:

$$s_i(t) = \max\left(0, \frac{E_i(0) - E_i(t)}{E_i(0)}\right). \quad (5.6)$$

Each agent has a baseline resistance threshold θ_i . Thresholds may be homogeneous or drawn from a distribution $P(\theta)$, for example a truncated normal distribution with mean μ_θ and dispersion σ_θ .

Local contagion is represented through an effective threshold. Let

$$f_i(t) = \begin{cases} \frac{1}{|\mathcal{N}_i(R)|} \sum_{j \in \mathcal{N}_i(R)} \mathbb{I}\{F_j(t) = 1\}, & |\mathcal{N}_i(R)| > 0, \\ 0, & |\mathcal{N}_i(R)| = 0. \end{cases} \quad (5.7)$$

be the share of failed neighbors. The second case is needed for the control setting $R = 0$, where

the agent has no local neighbors and no division by zero occurs. The effective threshold is

$$\theta_i^{\text{eff}}(t) = \max(\theta_{\min}, \theta_i[1 - \beta f_i(t)]), \quad (5.8)$$

where $\beta \in [0, 1]$ is the contagion parameter and $\theta_{\min}$ prevents the threshold from becoming unrealistically small.

Agent i fails when

$$s_i(t) \geq \theta_i^{\text{eff}}(t). \quad (5.9)$$

After failure, the agent liquidates its position through a market order and stops trading. The liquidation changes the order book and price, which can increase stress for other agents. At each iteration, the cascade update repeats until no new failures occur or a maximum number of internal rounds is reached.

5.4 Shock mechanisms

The simulator includes both single shocks and repeated shocks. The repeated-shock class applies the same base event at equally spaced iterations:

$$t_k = t_0 + k\Delta T, \quad k = 0, \dots, K - 1, \quad (5.10)$$

where t_0 is the first shock time, ΔT is the inter-shock step, and K is the number of shocks.

Three event families are used in the experiments. A repeated market-price shock shifts the order book by Δp in each event and is compared with a single benchmark shock of size $K\Delta p$. A repeated liquidity shock submits repeated market orders that consume one side of the book. A local cash shock affects a contiguous interval around center i_0 with integer width w :

$$S_0 = \{i \in \{1, \dots, N\} : |i - i_0| \leq \lfloor w/2 \rfloor\}. \quad (5.11)$$

A negative local cash shock reduces capital in S_0 and can trigger forced liquidations.

The extended experiment also uses relative price shocks. A one-time benchmark shock is a -20% market move. For repeated shocks of nominal amplitude $a\%$, the number of shocks is chosen as

$$K(a) = \left\lceil \frac{20}{a} \right\rceil, \quad (5.12)$$

so that the nominal total shock budget remains comparable to the benchmark. If percentage shocks are implemented multiplicatively, the realized cumulative price effect is

$$D(a, K) = 1 - \left(1 - \frac{a}{100}\right)^K, \quad (5.13)$$

which is generally not exactly equal to 20% . Therefore, the comparison is interpreted as a nominal-budget benchmark rather than as an exact equality of cumulative percentage price changes.

5.5 Outcome metrics

A cascade is identified when at least one agent outside the initially shocked interval fails:

$$\exists i \notin S_0 \text{ such that } t_{\text{fail},i} < +\infty. \quad (5.14)$$

The final failed fraction is

$$C = \frac{1}{N} \sum_{i=1}^N \mathbb{I}\{t_{\text{fail},i} < +\infty\}. \quad (5.15)$$

The outside-cascade fraction is

$$C_{\text{out}} = \frac{1}{N} \sum_{i \notin S_0} \mathbb{I}\{t_{\text{fail},i} < +\infty\}. \quad (5.16)$$

For local shocks, the spatial extent is

$$L = \max_{i \in \mathcal{F}} |i - i_0|, \quad (5.17)$$

where $\mathcal{F} = \{i : t_{\text{fail},i} < +\infty\}$ is the set of failed agents. The cumulative failed fraction over time is

$$C(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}\{t_{\text{fail},i} \leq t\}. \quad (5.18)$$

Price impact is summarized through maximum drawdown,

$$DD_{\text{max}} = \max_t \left(\frac{\max_{u \leq t} P_u - P_t}{\max_{u \leq t} P_u} \right), \quad (5.19)$$

and through recovery time. If P_{pre} is the average price over the 50 iterations before the first shock and t_K is the last shock time, recovery occurs at the first $t \geq t_K$ such that the price remains within a $\pm 2\%$ corridor for 10 consecutive periods:

$$t_{\text{rec}} = \min \left\{ t \geq t_K : \frac{|P_{t+j} - P_{\text{pre}}|}{P_{\text{pre}}} \leq 0.02, j = 0, \dots, 9 \right\}. \quad (5.20)$$

The recovery time is $T_{\text{rec}} = t_{\text{rec}} - t_K$. If the set in (5.20) is empty within the simulated horizon T_{max} , recovery is treated as censored and the recorded value is $T_{\text{max}} - t_K$. This convention keeps unrecovered runs in the analysis without incorrectly assigning them a finite recovery time.

To measure executable liquidity beyond the best bid and ask, the fixed-volume spread is also used. For an evaluation volume Q_L , let $P_{\text{ask},t}^{\text{VWAP}}(Q_L)$ be the volume-weighted average purchase price across the best ask levels needed to fill Q_L , and define $P_{\text{bid},t}^{\text{VWAP}}(Q_L)$ analogously. Then

$$S_t^{\text{FV}}(Q_L) = P_{\text{ask},t}^{\text{VWAP}}(Q_L) - P_{\text{bid},t}^{\text{VWAP}}(Q_L). \quad (5.21)$$

If either side of the book lacks sufficient depth, the value is treated as undefined rather than

extrapolated.

6 Simulation Design and Dataset

The dataset is generated by simulations rather than by external historical observations. Each observation is a recorded market state at one iteration of the simulator. The logger stores prices, spreads, order-book quantities, agent equity, cash, assets, failure states, stresses, effective thresholds, and failure times. Run-level outcomes are then computed from these time series.

The empirical work is organized into three complementary blocks. The first block uses representative absolute price shocks and local cash shocks to visualize single-run dynamics and spatial patterns of failure propagation. The second block uses sensitivity plots over interaction radius, contagion strength, and mean resistance thresholds. The third block performs a frequency scan for repeated relative price shocks and compares them with a single -20% benchmark shock.

The absolute-shock block compares a repeated market-price shock of five events to a single benchmark shock with the same total nominal price change. The representative setting uses $t_0 = 100$, $K = 5$, $\Delta T = 20$, and $\Delta p = -5$, while the benchmark uses one shock of size -25 at t_0 . These figures are used as mechanism diagnostics rather than as final hypothesis tests.

The controlled local-cash validation uses a homogeneous baseline with repeated `CascadingLocalCashShock` events. The baseline configuration has $T = 700$, $t_0 = 100$, $K = 5$, $\Delta T = 20$, local cash change -100 , shock width $w = 5$, central shock position $N/2$, $\mu_\theta = 0.25$, $\sigma_\theta = 0.05$, interaction radius $R = 1$, contagion strength $\beta = 0.5$, and one market maker. The threshold sweep varies $\mu_\theta \in \{0.15, 0.25, 0.35, 0.45\}$. The formal validation of the local-cash block uses 100 fixed random seeds.

The relative-shock block uses a single -20% benchmark shock and repeated shocks with amplitudes $a \in \{0.5, 1, 2, 3, 5\}\%$. The inter-shock steps are selected from $\{1, 2, 3, 5, 8, 10, 15, 20, 30, 40\}$. For each amplitude, $K(a) = \lceil 20/a \rceil$ is used to keep the nominal shock budget close to the benchmark. Since repeated percentage shocks may compound multiplicatively, this comparison is interpreted as a nominal-budget comparison, not as exact equality of cumulative price change. The statistical validation of this block uses 30 fixed random seeds.

Table 6.1 summarizes the main controlled factors.

Table 6.2 gives a compact reproducibility summary for the two main validation blocks.

The statistical validation uses run-level outcomes. For numerical metrics, mean differences are accompanied by bootstrap confidence intervals and nonparametric or paired tests where appropriate. For binary cascade indicators, the validation reports risk differences and exact or chi-square tests. Families of pairwise comparisons are corrected using the Holm–Bonferroni procedure. Graphical diagnostics are used to interpret mechanisms, while the final hypothesis decisions are based on the formal validation in Section 8.

Table 6.1: Main experimental factors and interpretation.

Factor	Values or definition	Purpose
Shock timing	Single benchmark vs. repeated shocks with step ΔT	Tests whether timing matters beyond total nominal magnitude
Shock channel	Market price, liquidity, local cash, relative price	Separates price-level shocks from localized balance-sheet shocks
Interaction radius	$R = 0, 1, 2, 3, 4, 5$ in visual sensitivity runs; $R = 0$ vs. $R = 1$ in the local-cash ablation	Tests locality of propagation
Contagion strength	$\beta \in [0, 1]$ in visual sensitivity runs; $\beta = 0$ vs. $\beta = 0.5$ in the local-cash ablation	Tests threshold reduction through failed neighbors
Threshold mean	$\mu_\theta \in \{0.15, 0.25, 0.35, 0.45\}$ in statistical validation	Tests agent-level resistance
Random seeds	100 for local-cash validation; 30 for relative-shock validation	Measures stochastic robustness

Table 6.2: Core simulation protocol and parameter values.

Parameter	Local-cash controlled validation	Relative-shock frequency scan
Validation seeds	100 seeds, indexed $1, \dots, 100$	30 seeds, indexed $1, \dots, 30$
Iterations	$T = 700$	Scenario-specific horizon
Initial shock time	$t_0 = 100$	$t_0 = 100$
Repeated shocks	$K = 5, \Delta T = 20$	$K(a) = \lceil 20/a \rceil$
Shock amplitude	Local cash change -100	$a \in \{0.5, 1, 2, 3, 5\}\%$
Local shock width	$w = 5$, center $N/2$	Not local; price-level shock
Baseline cascade parameters	$R = 1, \beta = 0.5$	$R = 1, \beta = 0.5$
Propagation-suppressed control	$R = 0, \beta = 0$	Not applicable
Thresholds	$\mu_\theta = 0.25, \sigma_\theta = 0.05$ in baseline; μ_θ swept for H3	Same baseline threshold layer
Bootstrap validation	5000 resamples, fixed bootstrap seed	5000 resamples, fixed bootstrap seed

7 Empirical Results

7.1 Repeated market shocks versus the single benchmark

Figures 7.1 and 7.2 compare a sequence of five absolute price shocks with a single benchmark shock of equal total nominal price change. The benchmark produces an immediate sharp price drop at $t = 100$, followed by partial recovery. The repeated-shock scenario distributes the disturbance over $t = 100, 120, 140, 160, 180$ and therefore keeps the market under stress for a longer interval. This comparison is used as an illustrative diagnostic: it shows that shock timing changes the failure trajectory, but the exact number of failed traders depends on seed and configuration and is not used as the final hypothesis test.

The seed-level comparison in Figures 7.3 and 7.4 shows why the result should not be interpreted deterministically. The repeated-shock scenario exceeds the benchmark for some seeds but not for all. This indicates that repeated shocks change the distribution of risk rather than guaranteeing a larger cascade in every run.

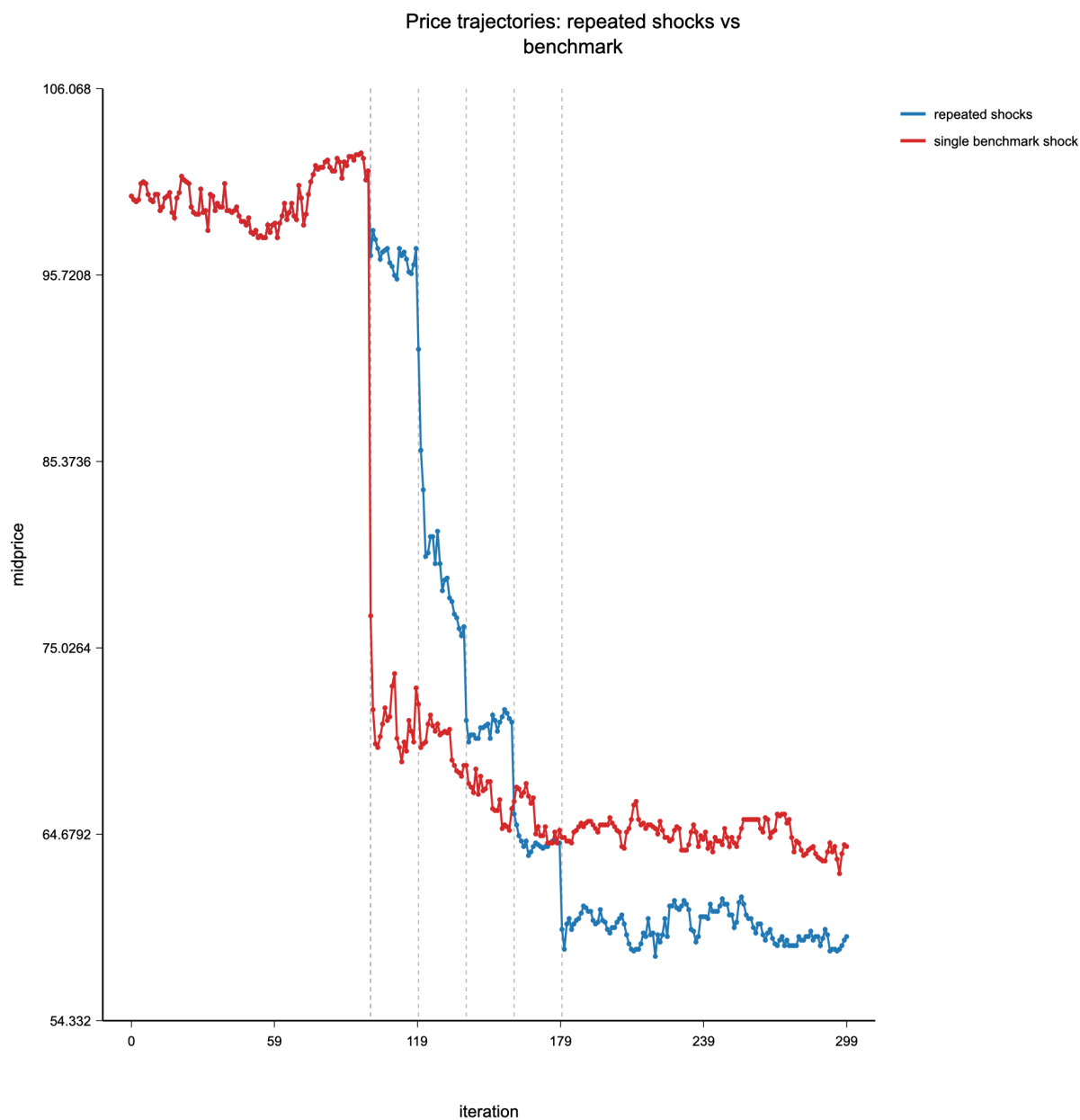


Figure 7.1: Price trajectories in an illustrative comparison between a benchmark single price shock and a sequence of smaller shocks with the same total nominal price change. Dashed vertical lines indicate shock times in the repeated-shock scenario.

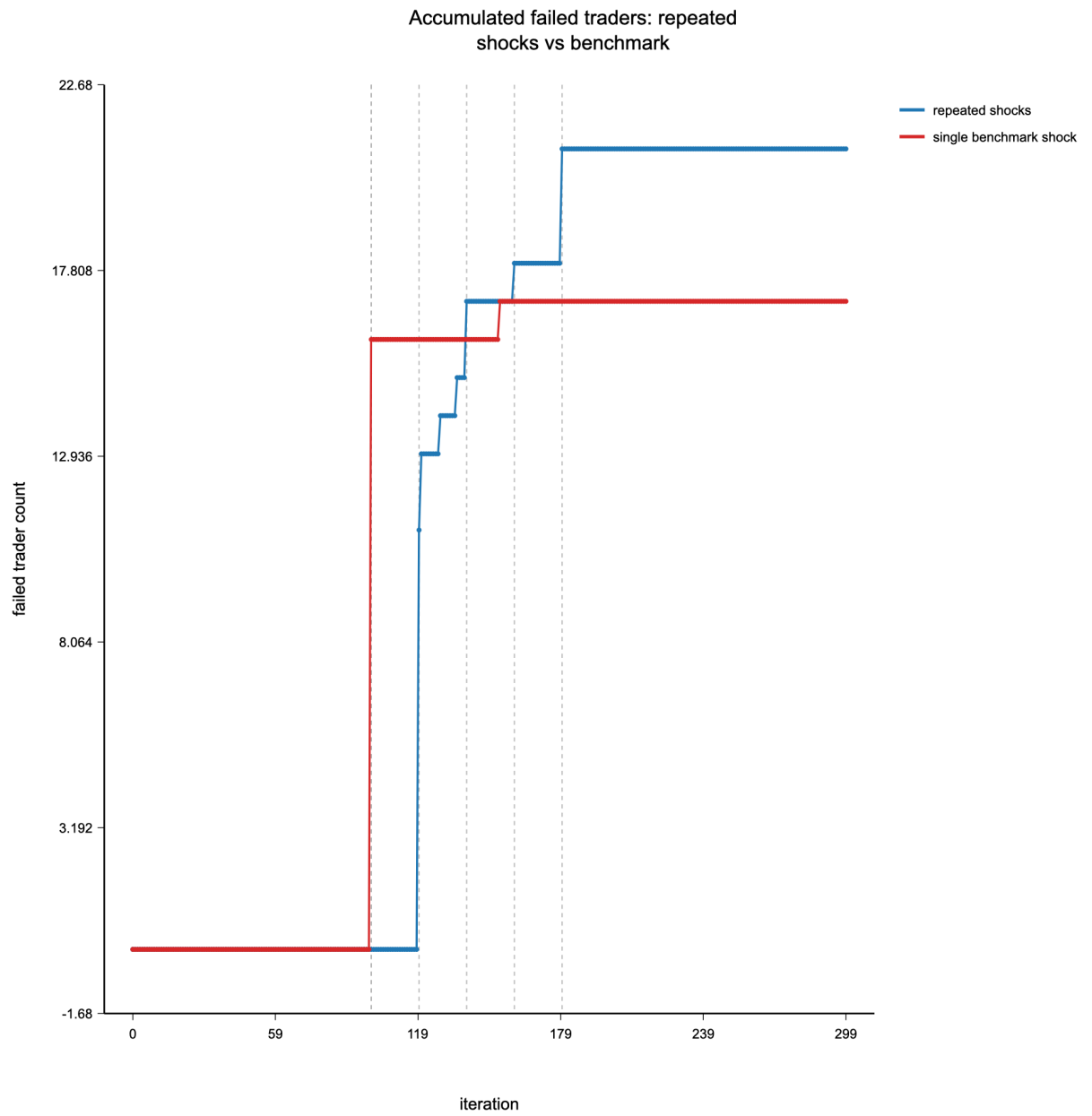


Figure 7.2: Accumulated failed traders in the same illustrative benchmark-versus-repeated-shock comparison.

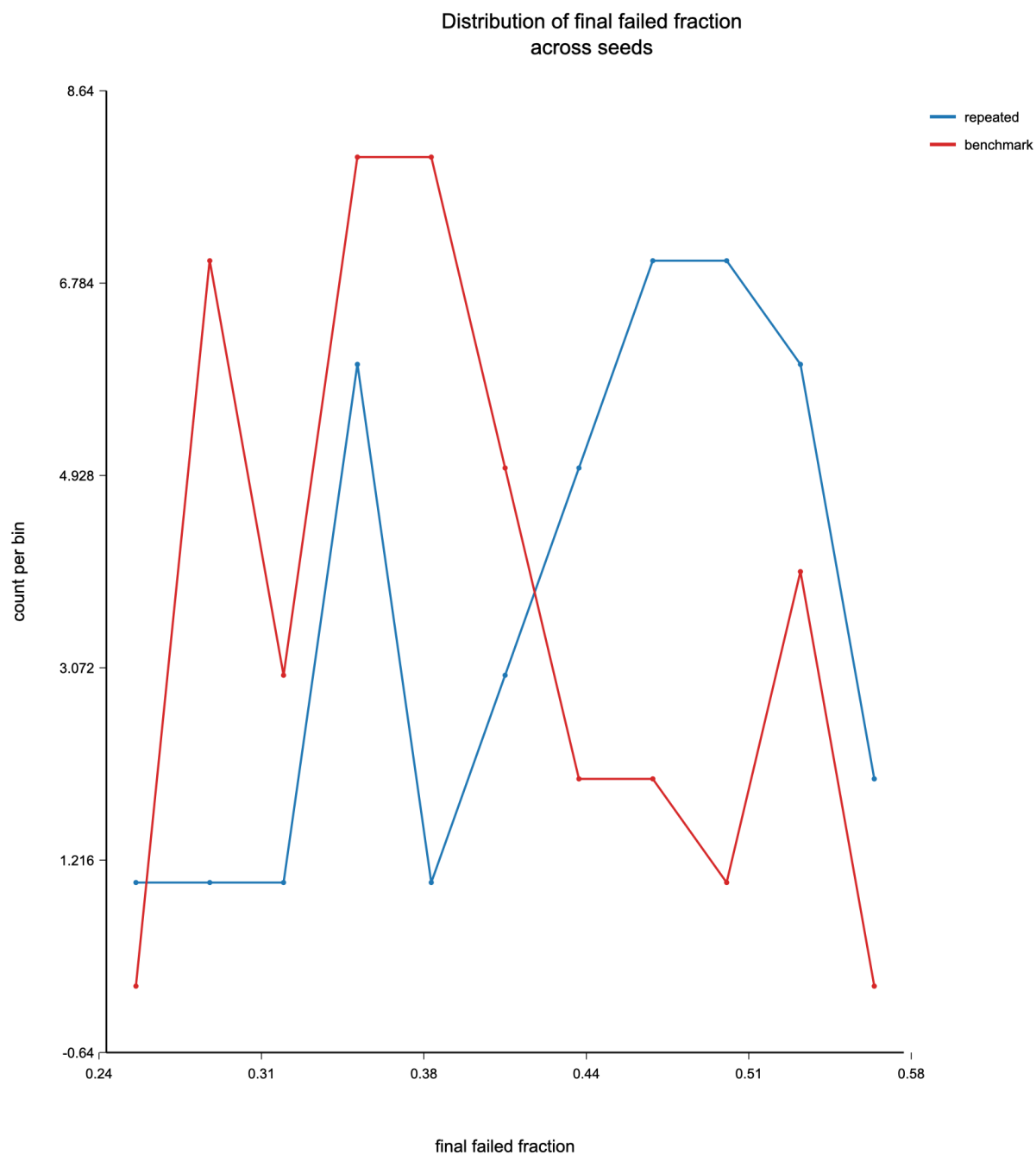


Figure 7.3: Distribution of final failed fraction across seeds for the benchmark and repeated absolute-shock scenarios.

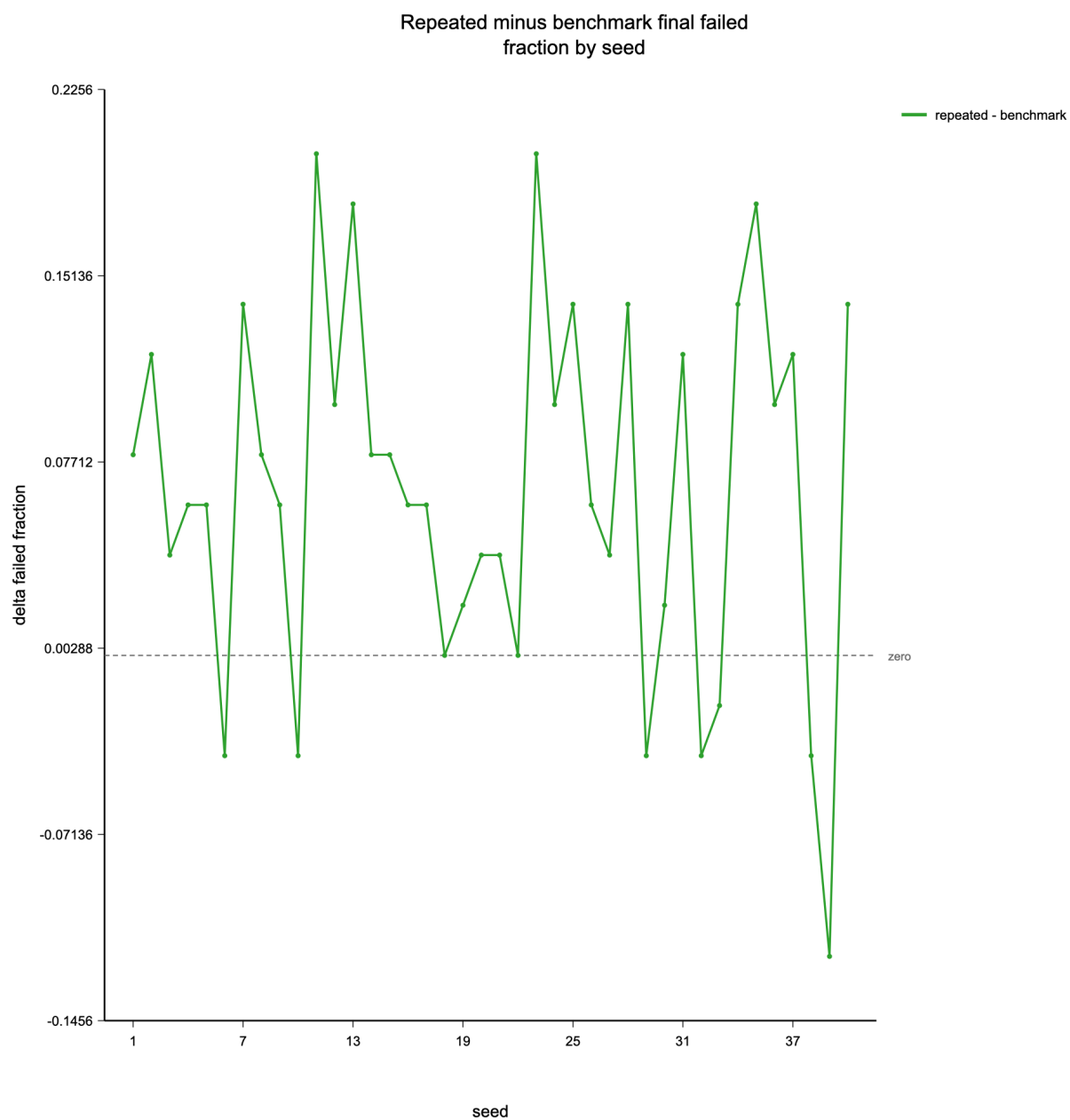


Figure 7.4: Seed-by-seed difference in final failed fraction between the repeated-shock and benchmark runs. Positive values mean that the repeated-shock run produced a larger failed fraction for the same seed.

7.2 Spatial diagnostics under cascade and local cash shocks

Figures 7.5 and 7.6 show the space-time structure of failures and stress in one repeated-shock scenario. Failures appear as persistent regions in the failure heatmap, while the stress heatmap shows where capital drawdowns accumulate before and during failures. These figures illustrate how the implemented cascade layer can generate spatially structured failure patterns.

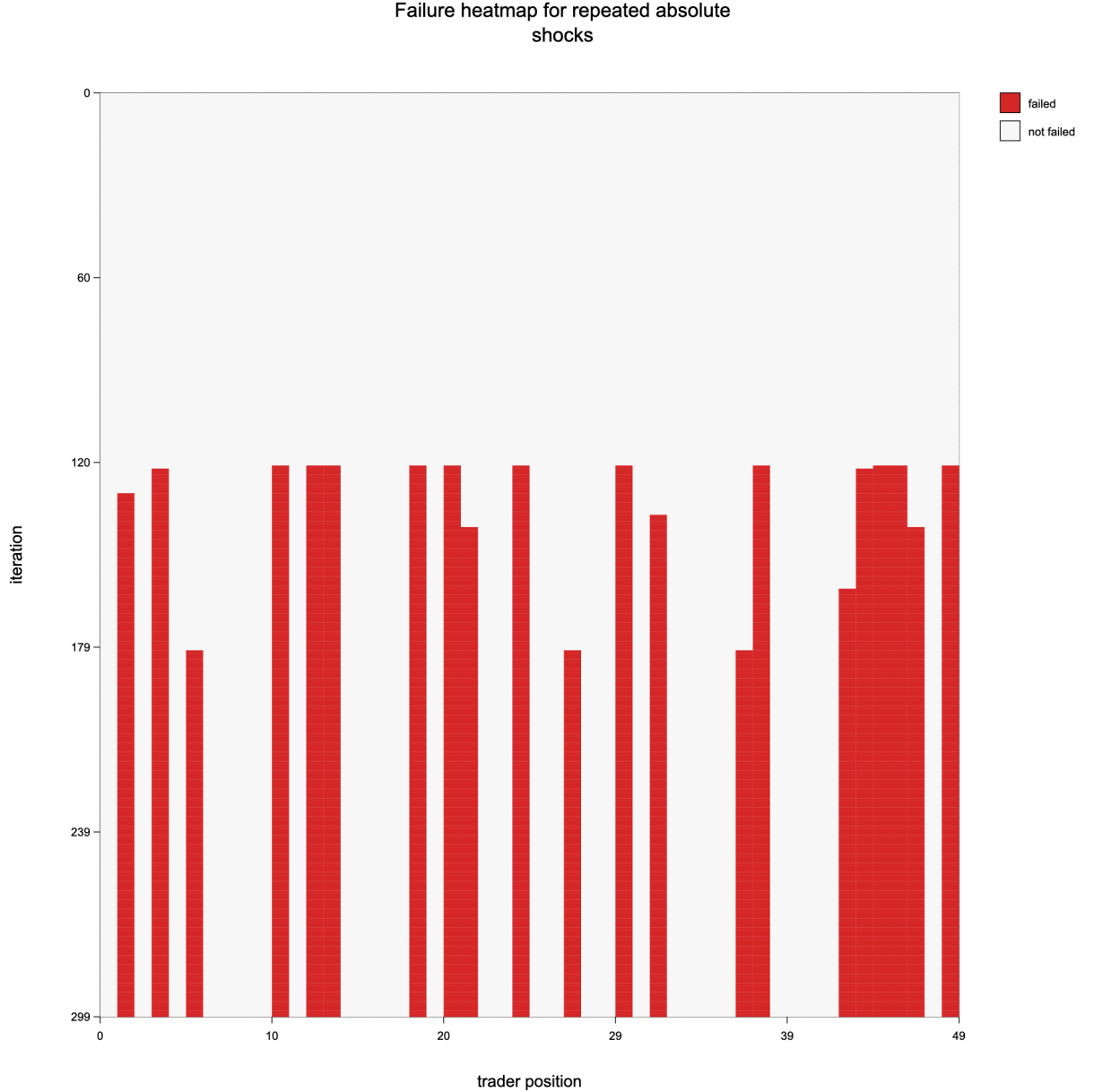


Figure 7.5: Failure state over time and position in one repeated-shock cascade scenario.

The local-cash-shock example in Figures 7.7 and 7.8 is useful for visualizing the distinction between the initially affected interval and the final failure set. In this illustrative run, failures spread beyond the local shock region. However, the formal ablation test in Section 8 shows that this visual pattern is not statistically stronger in the propagation-enabled baseline than in the propagation-suppressed control under the final validation setup. Therefore, the heatmap is interpreted as a mechanism illustration rather than as final evidence for H1.

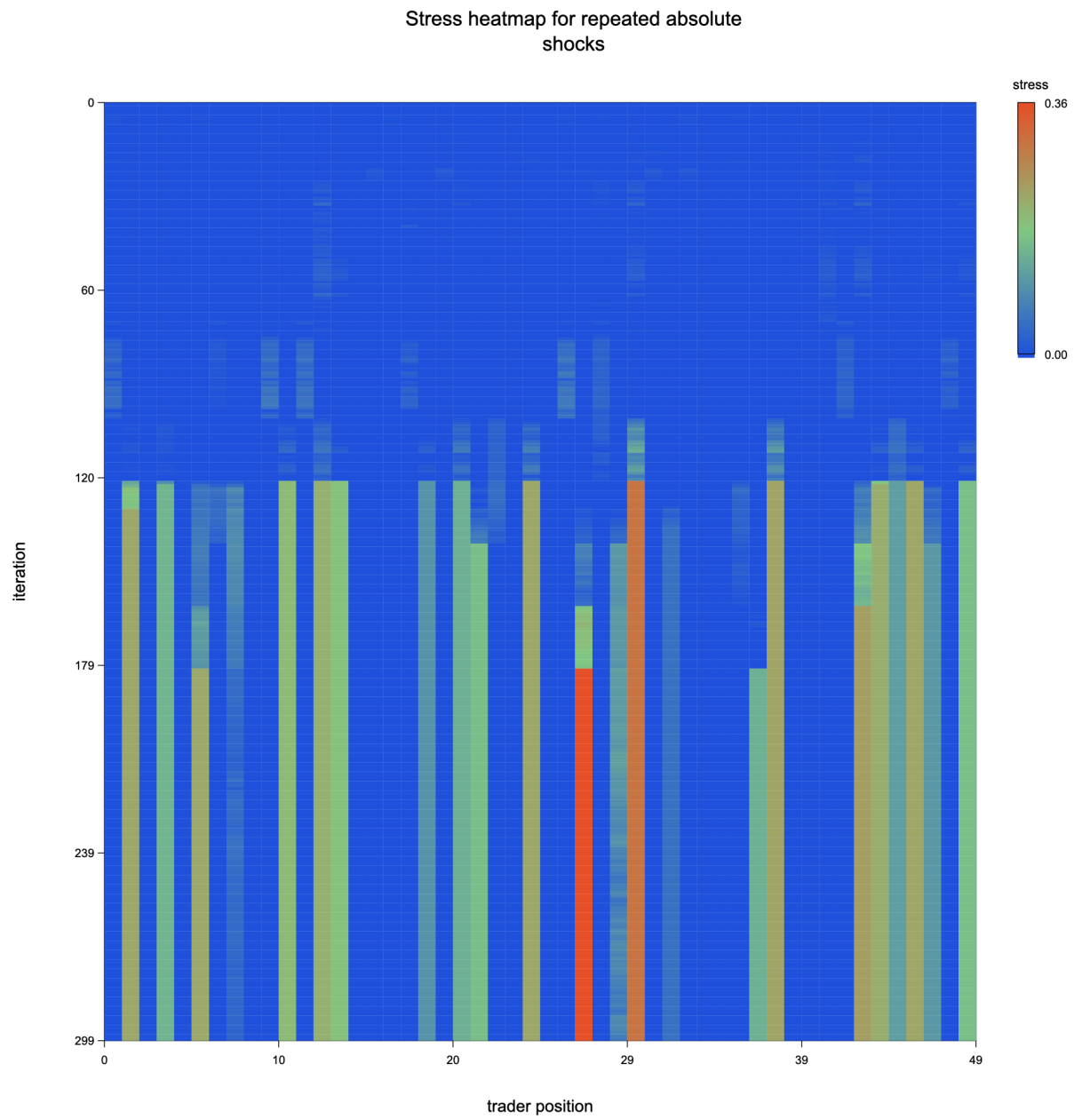


Figure 7.6: Stress over time and position in the same repeated-shock cascade scenario.

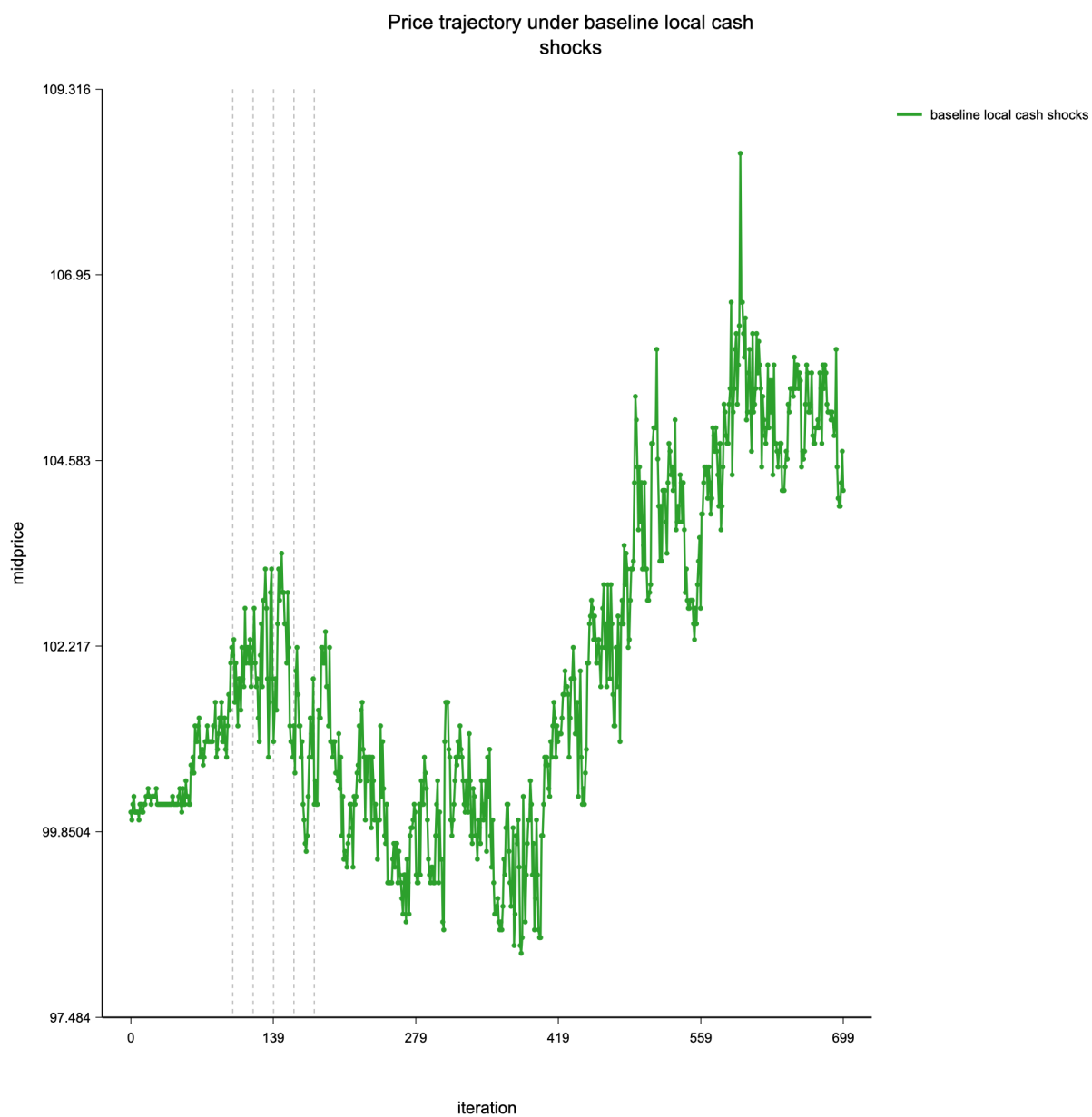


Figure 7.7: Price trajectory in an illustrative local cash-shock run.

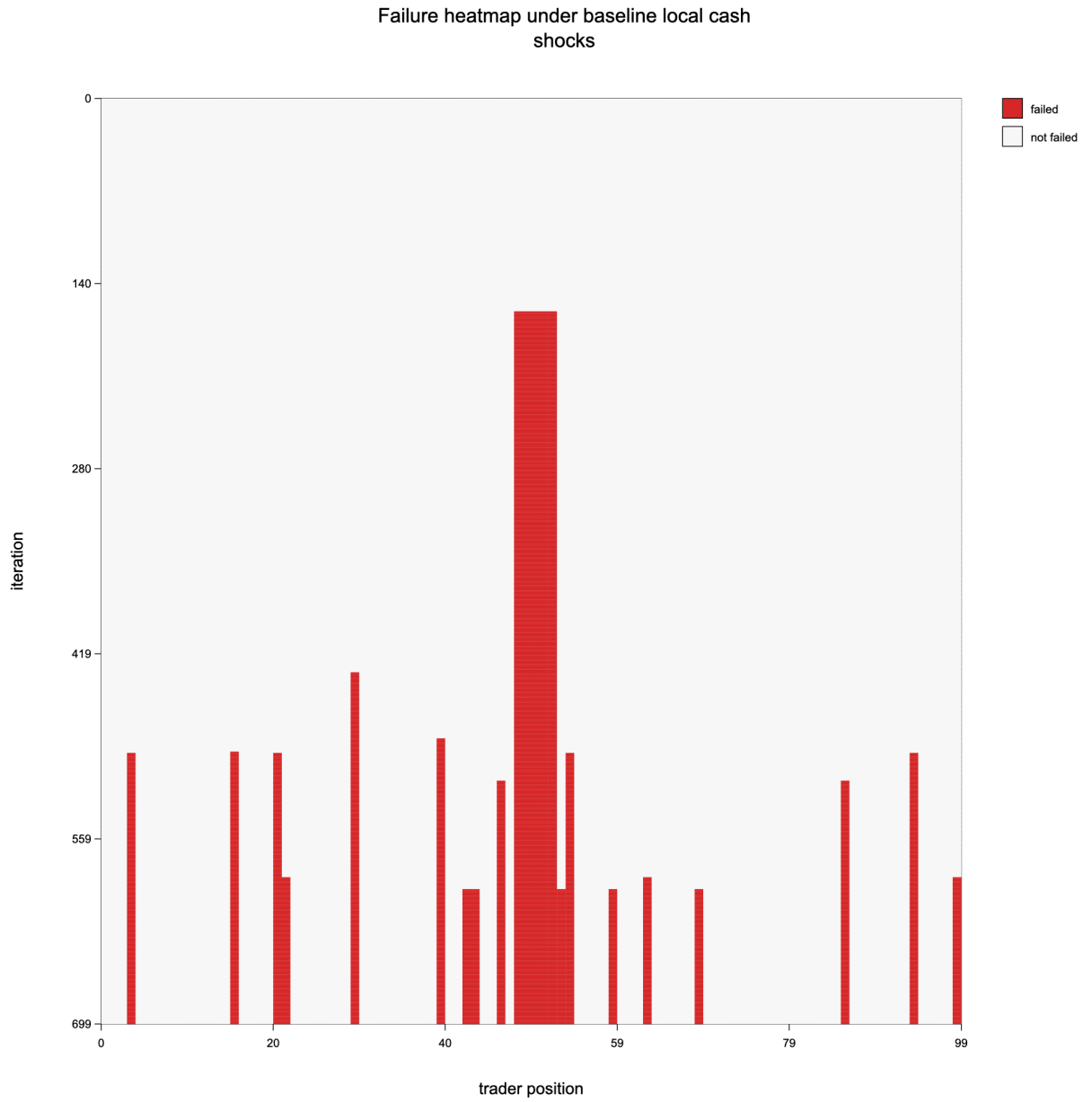


Figure 7.8: Failure propagation in the same illustrative local cash-shock run. The formal decision on localized propagation is based on the controlled validation in Section 8.

7.3 Sensitivity to interaction radius, contagion, and thresholds

The sensitivity figures are generated from the final codebase and are used to interpret how local interaction parameters affect simulated failures. Figure 7.9 reports the dependence of two run-averaged cascade metrics on the interaction radius: final cascade size and cascade occurrence. The relationship is not monotone in the reported range: the run-averaged cascade size changes only moderately, while the cascade indicator decreases from $R = 0$ to $R = 1$ and then partially rebounds. This suggests that wider neighborhoods do not automatically imply stronger cascades; in some regimes, additional neighbors may dilute local stress or make outcomes more seed-dependent.

Figure 7.10 shows the effect of the contagion parameter β on the same two mean cascade metrics. The response is non-monotone: the cascade indicator falls sharply at low positive β and then increases again at larger values. Since this figure does not display confidence intervals and the final statistical pipeline does not include an inferential β sweep, it is interpreted as a descriptive diagnostic rather than as formal validation of H3b.

Figure 7.11 provides the most systematic visual pattern. Higher mean resistance thresholds generally reduce both cascade probability and mean failed fraction, although the relation is not strictly monotone at the highest threshold level. This visual result is consistent with the formal threshold validation in Section 8.

Figure 7.12 plots heterogeneous thresholds and marks failed agents in one representative local-cash run. The plot illustrates that failures are not determined only by the lowest baseline thresholds; realized vulnerability depends on price movements, capital drawdown, and the dynamic failure process.

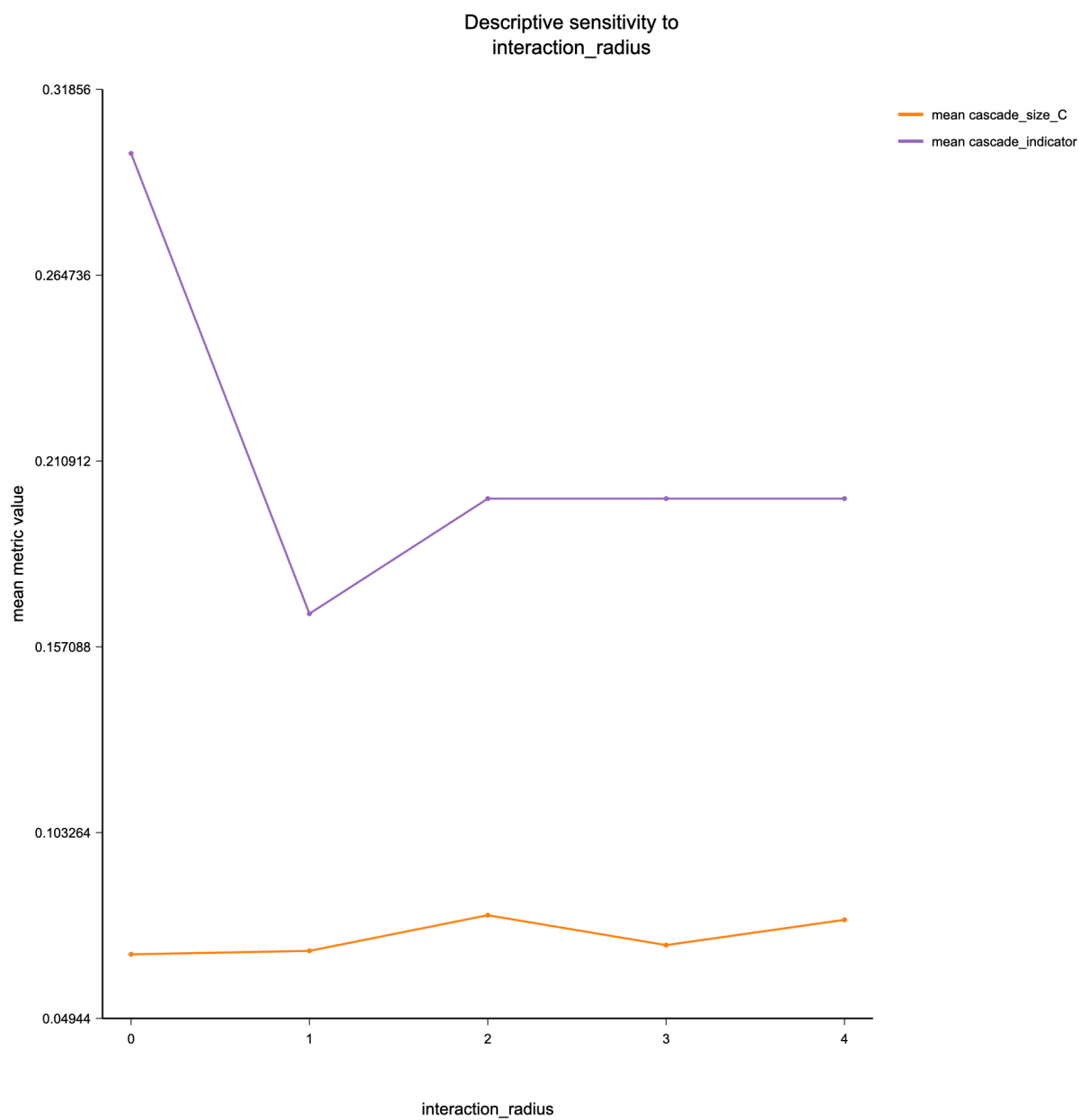


Figure 7.9: Descriptive sensitivity to interaction radius. The figure reports run-averaged cascade size and cascade indicator for the local-cash scenario; uncertainty intervals are not shown in this diagnostic plot.

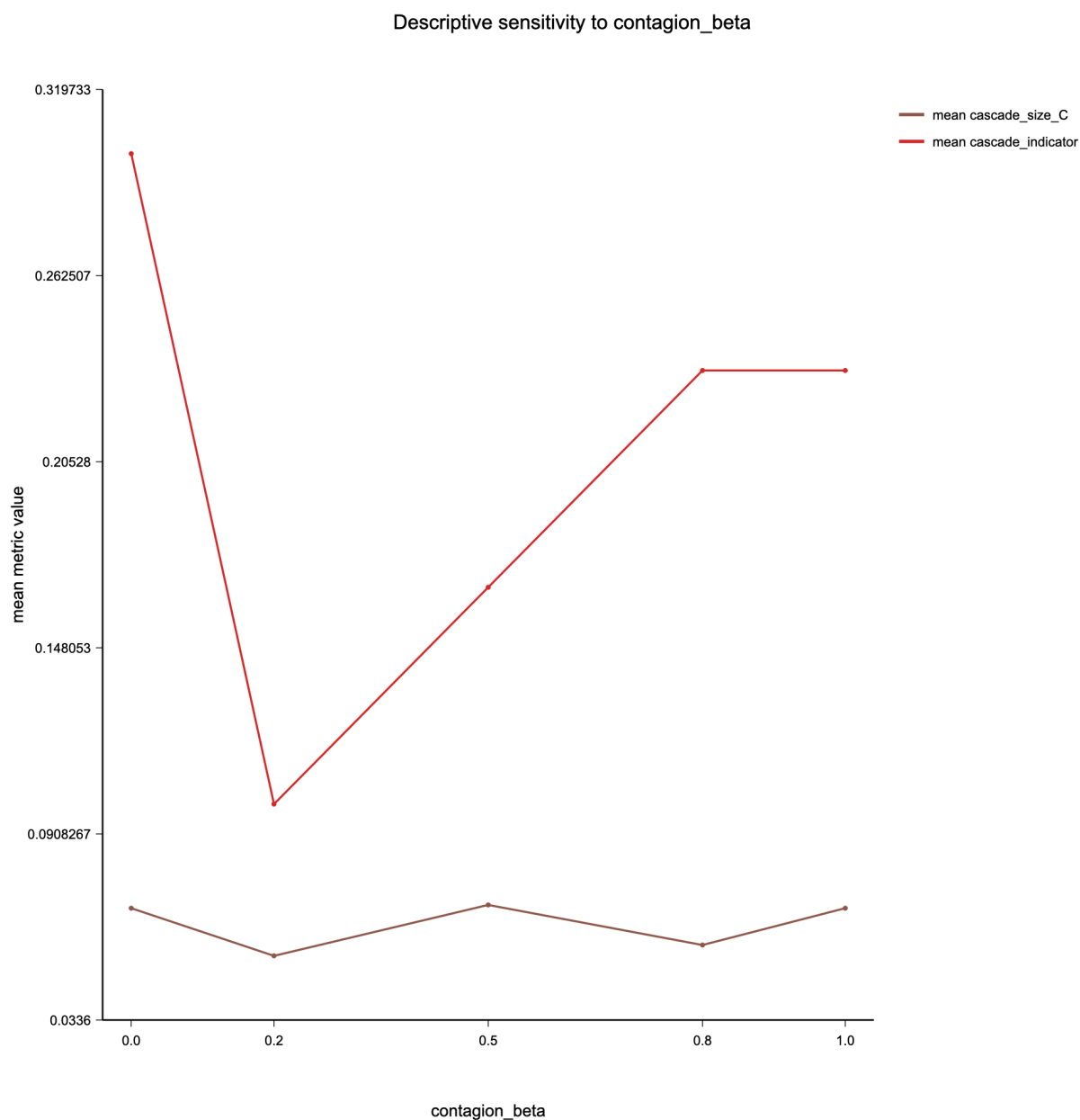


Figure 7.10: Descriptive sensitivity to contagion strength β . The figure reports run-averaged cascade size and cascade indicator; it is not used as an inferential test because the final statistical pipeline does not implement a controlled β sweep.

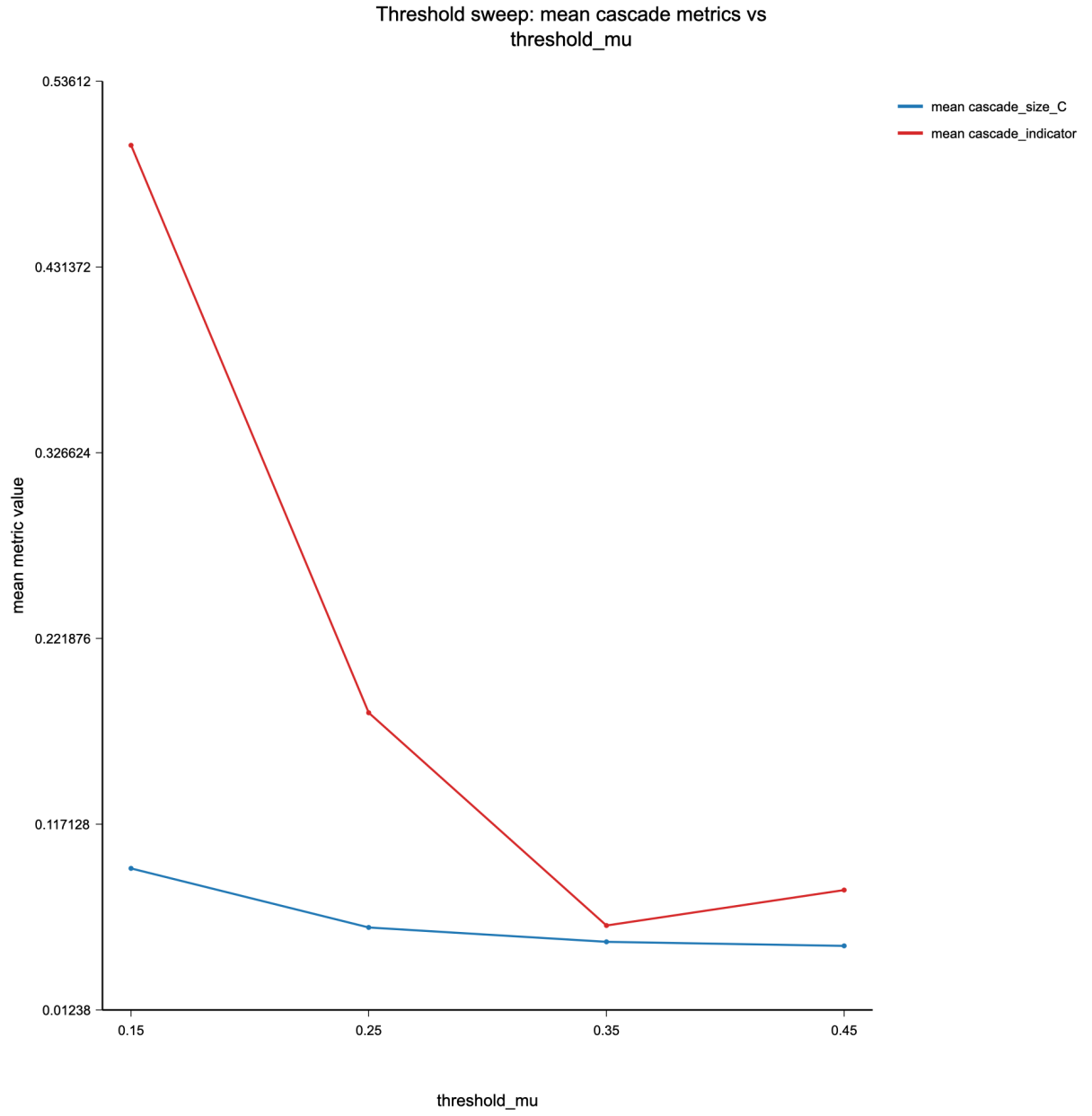


Figure 7.11: Sensitivity to the mean resistance threshold μ_θ . This is the main visual diagnostic for H3a: higher thresholds generally reduce cascade frequency and size, although the response is not perfectly monotone at every threshold level.

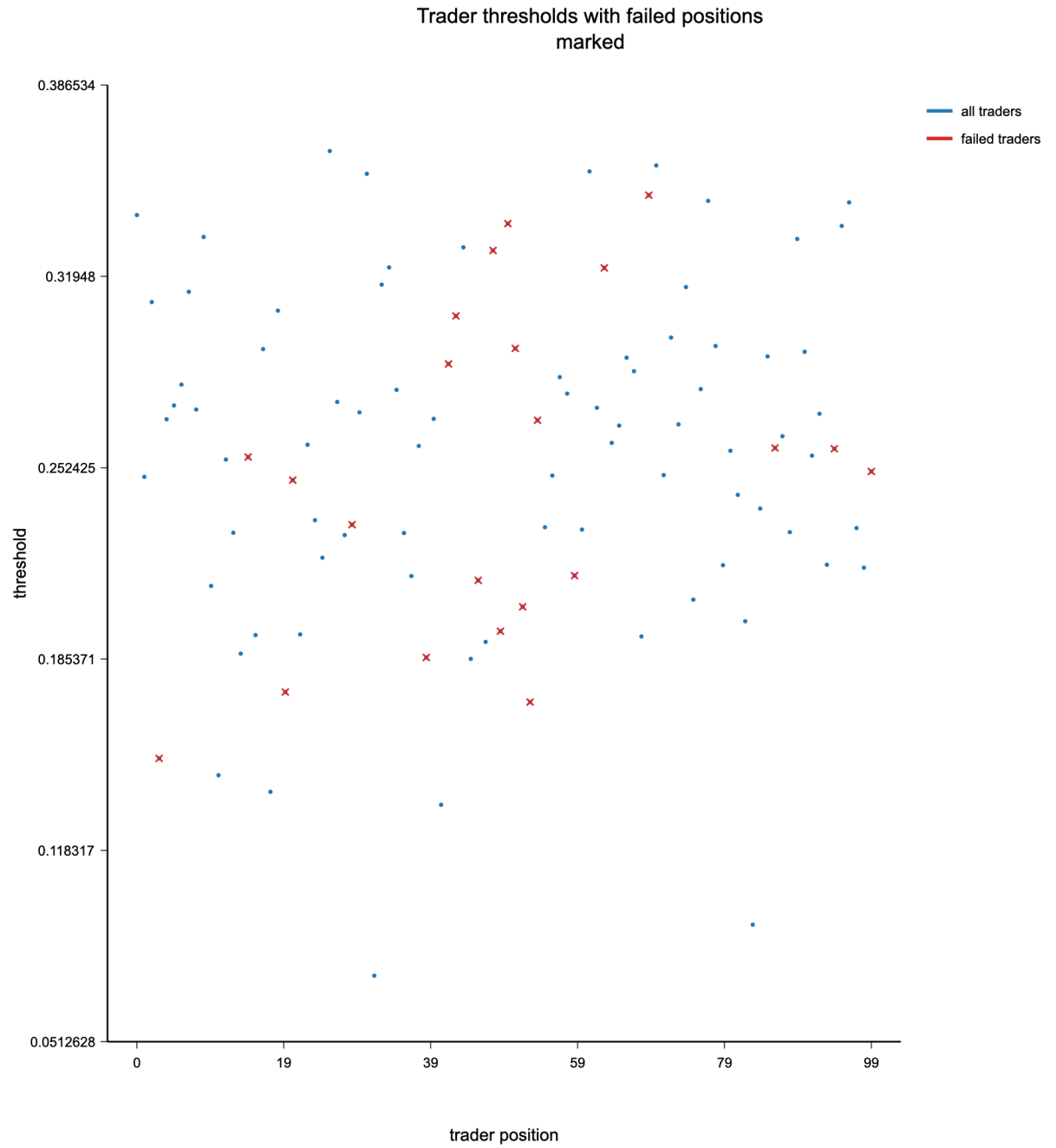


Figure 7.12: Heterogeneous thresholds and failed agents in one representative run. Failed agents are marked by crosses.

7.4 Frequency scan for repeated relative shocks

The relative-shock experiments provide the main evidence for H2. Figure 7.13 compares a single -20% benchmark shock with repeated 1% shocks at different inter-shock steps. When shocks arrive frequently, the repeated regime can generate price damage comparable to the benchmark. When the step is large, the response can be weaker or delayed. The comparison is nominal rather than exactly additive because repeated percentage shocks compound over time.

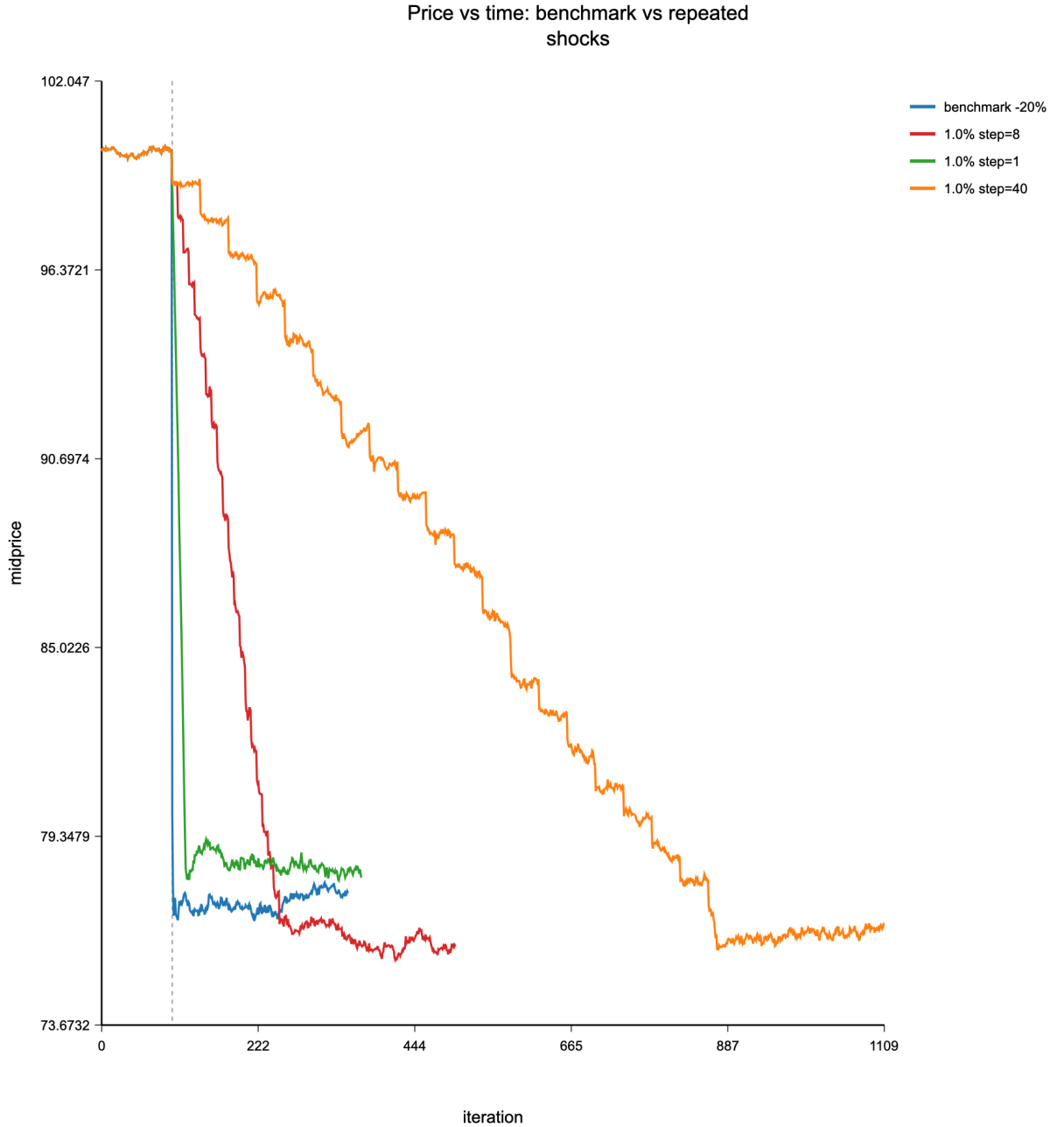


Figure 7.13: Price dynamics for the benchmark -20% shock and repeated 1% shocks at different inter-shock steps.

Figure 7.14 reports the corresponding failed fraction. The plot illustrates why repeated-shock risk cannot be reduced to total nominal shock size alone: the same nominal shock budget can generate different agent-level damage depending on timing.

Figures 7.15 and 7.16 summarize the amplitude-step grid. The heatmaps show that benchmark-

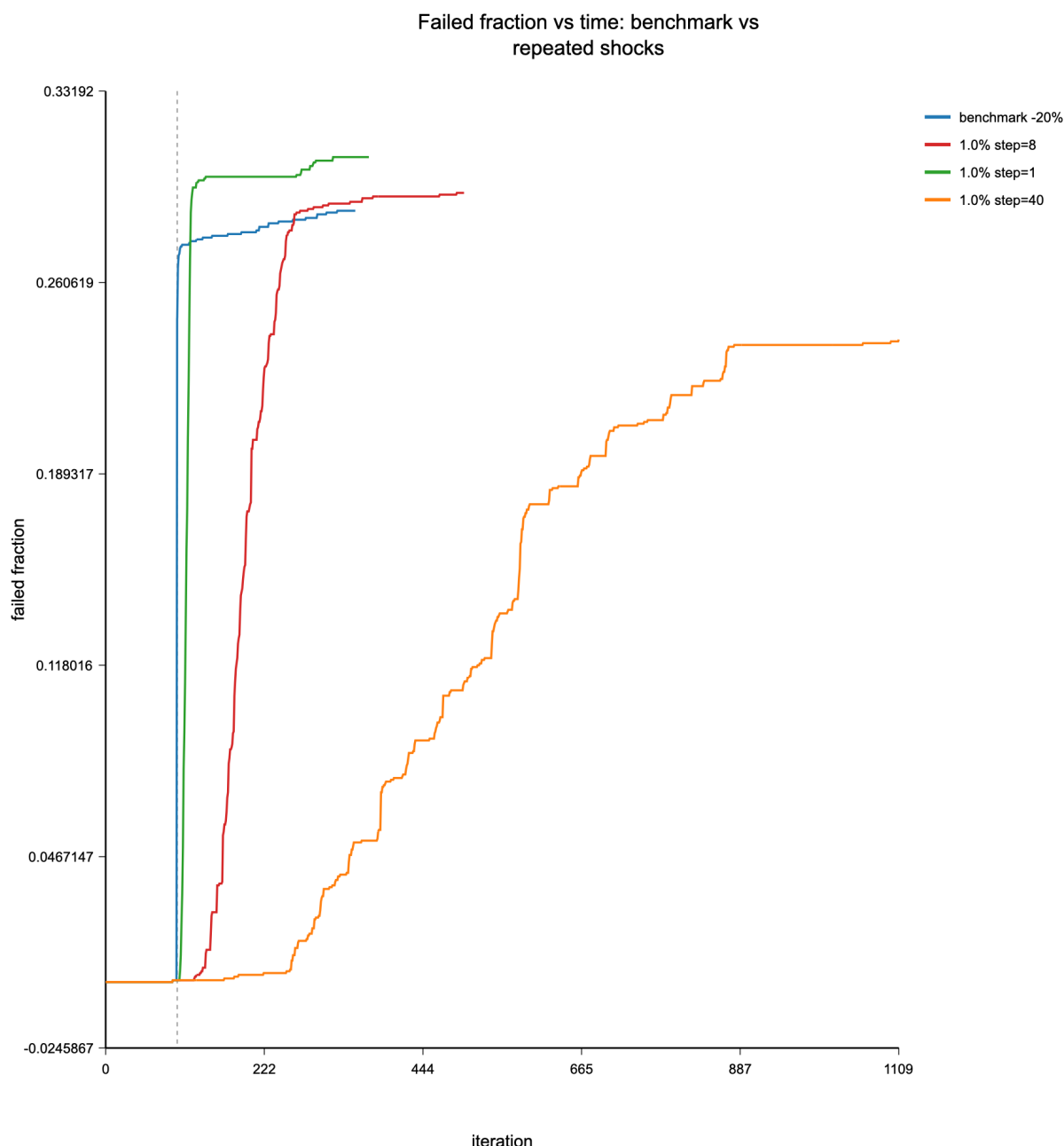


Figure 7.14: Failed fraction under the benchmark and repeated 1% relative-shock regimes. The vertical axis reports the share of failed agents.

like outcomes are not concentrated in a single scenario. Some amplitudes display monotone benchmark-like regions, whereas others display separated islands of comparability.

Figure 7.17 visualizes the binary benchmark-like classification. Red cells indicate regimes that satisfy the pre-specified benchmark-like tolerance rule: either the mean maximum drawdown is within 1 percentage point of the benchmark or the mean peak failed fraction is within 0.02 of the benchmark. This rule is an equivalence-style descriptive criterion; it is not a formal TOST equivalence test.

Table 7.1 reports the final critical-frequency classification obtained from the regenerated resonance scan. The 2%, 3%, and 5% regimes have monotone-prefix regions. The 0.5% and 1% regimes have separated islands of comparability, so they do not admit one clean critical step.

Figure 7.18 summarizes the same classification graphically using the regenerated archive

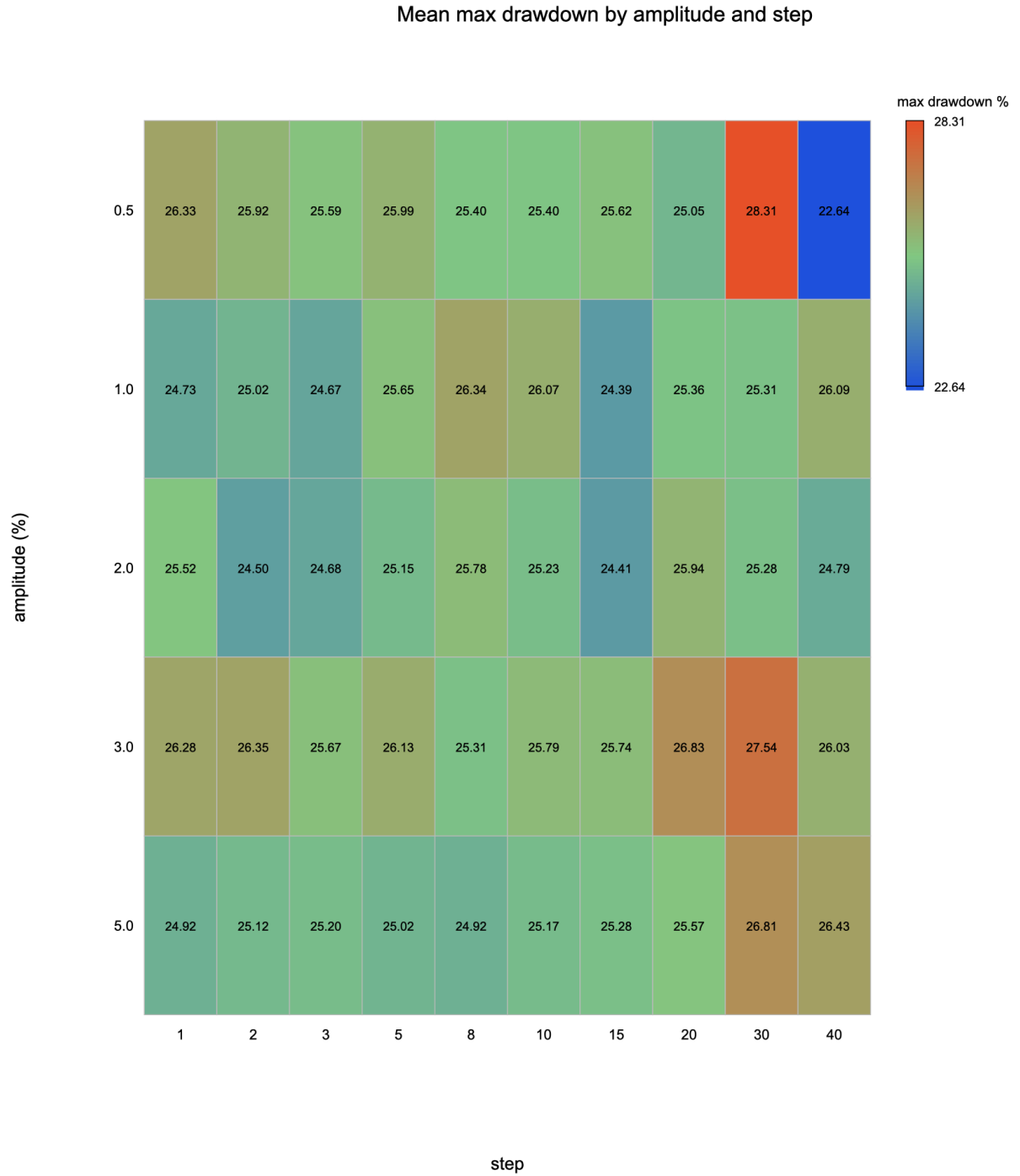


Figure 7.15: Mean maximum drawdown across the amplitude-step grid. Rows correspond to repeated-shock amplitudes and columns to inter-shock steps.

Table 7.1: Critical-frequency classification for repeated relative shocks.

Amplitude	Number of shocks	Critical step	Critical frequency	Pattern
0.5%	40	—	—	Islands: steps 1–15 and 30
1.0%	20	—	—	Islands: steps 1–10 and 20–40
2.0%	10	30	0.0333	Monotone prefix: comparable up to step 30
3.0%	7	40	0.0250	Monotone prefix: comparable up to step 40
5.0%	4	40	0.0250	Monotone prefix: comparable up to step 40

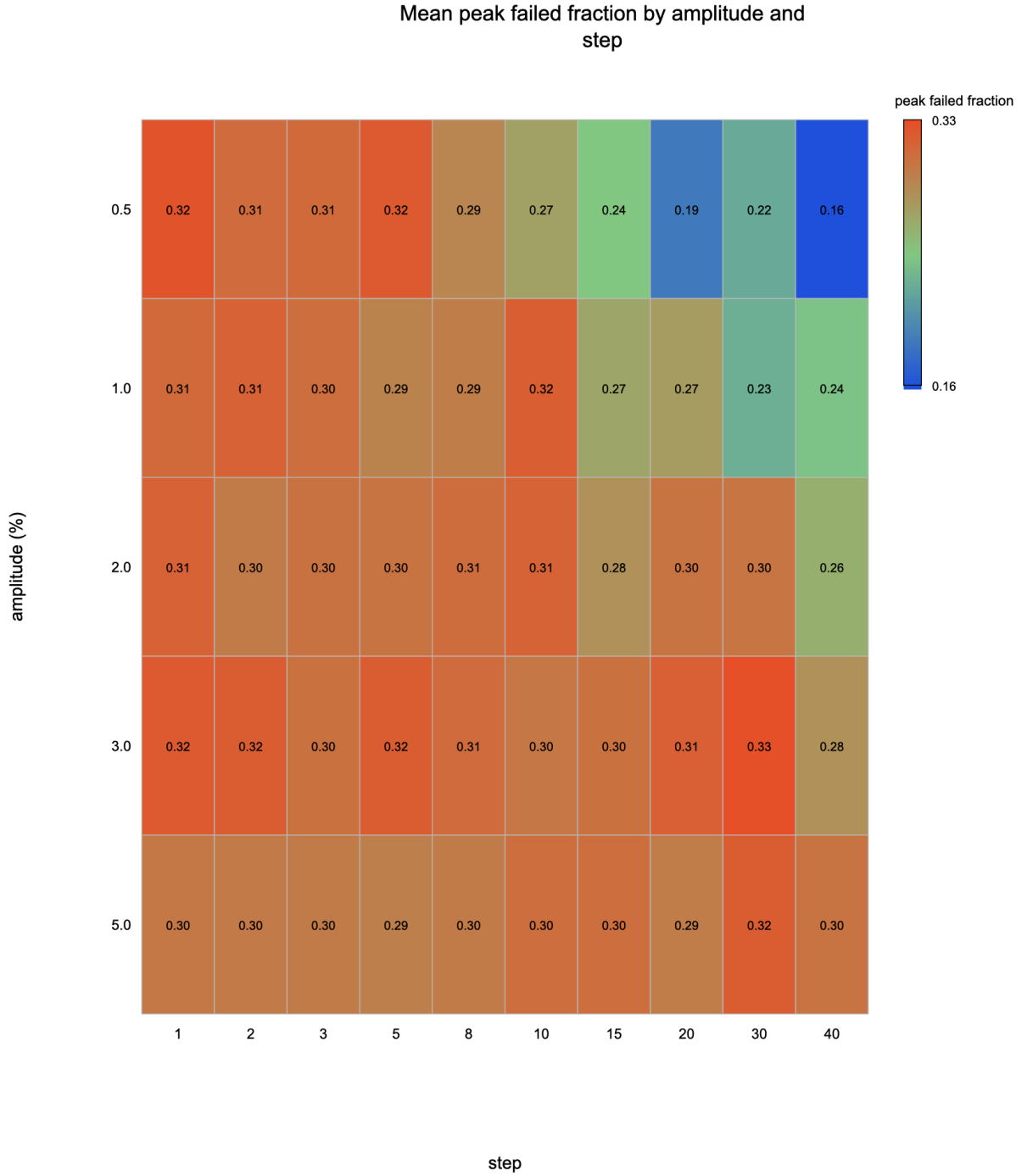


Figure 7.16: Mean peak failed fraction across the same amplitude-step grid. The color scale reports the peak share of failed agents.

figure. Gray points show all benchmark-like frequency–amplitude combinations, while the red line reports critical frequencies only for amplitudes with a monotone-prefix classification. Amplitudes with island patterns are therefore discussed in Table 7.1 rather than assigned a single plotted critical frequency.

The detailed 1% scan in Figure 7.19 illustrates why H2 is only partially supported. Several 1% regimes are benchmark-like, but the response is not monotone in the inter-shock step. In the full grid, the closest 1% regime by mean drawdown is step 8, with mean maximum drawdown of approximately 26.34% against the benchmark mean of 26.20%. This supports benchmark-like comparability, but not strict dominance or formal equivalence.

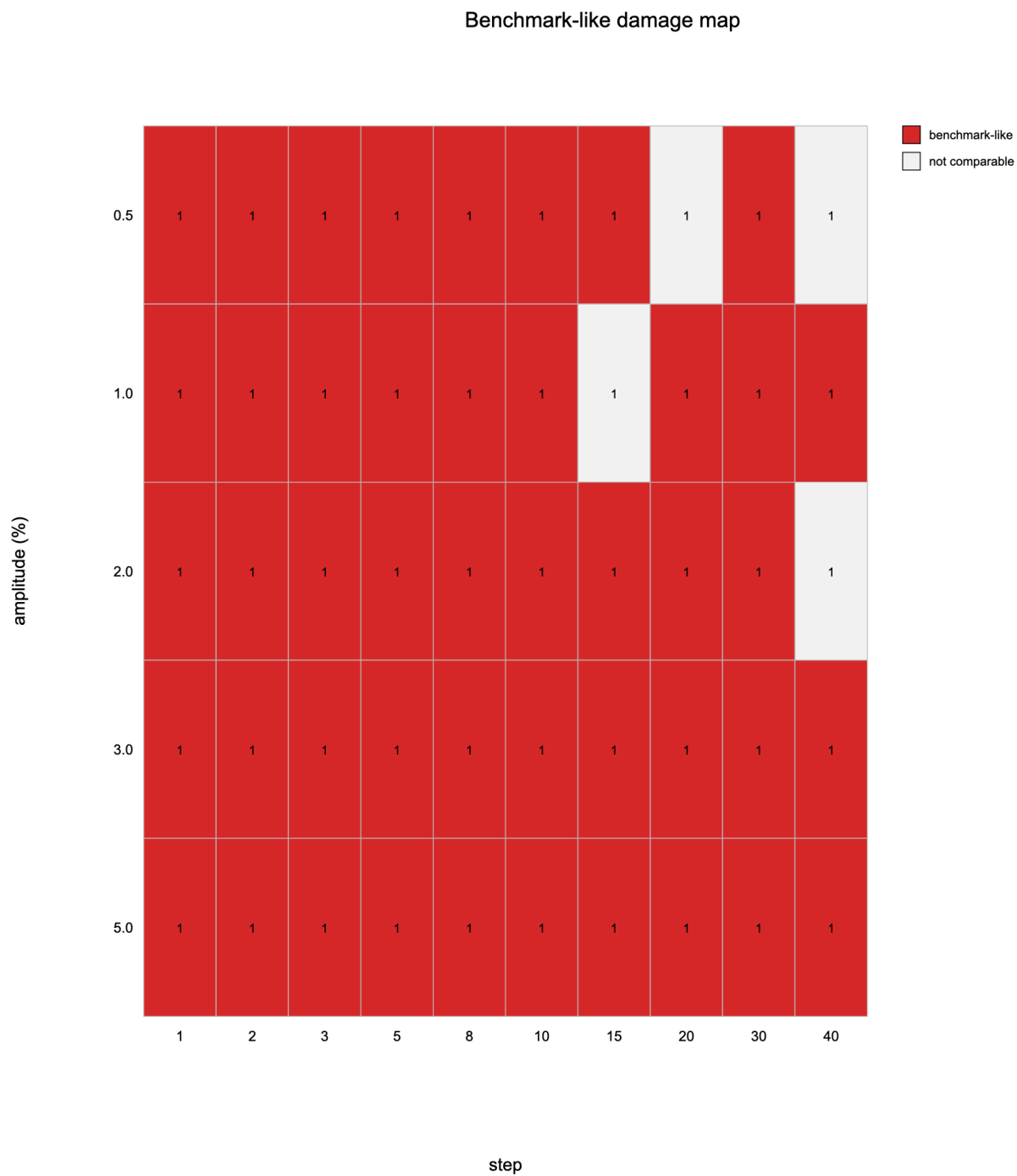


Figure 7.17: Benchmark-like damage map for repeated relative shocks. Red cells satisfy the pre-specified benchmark-like tolerance rule.

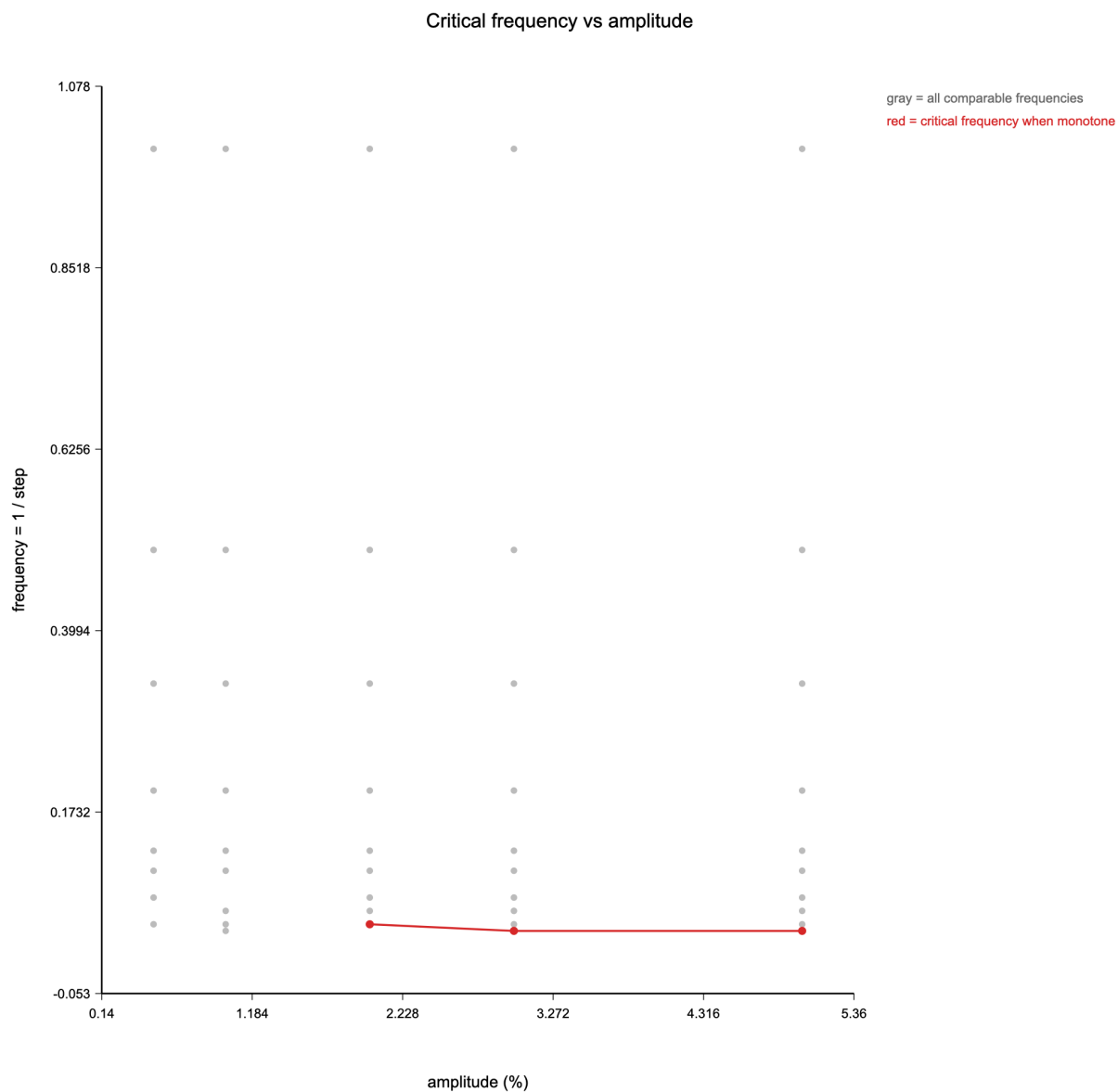


Figure 7.18: Critical-frequency summary for repeated relative shocks. Gray points show all benchmark-like regimes; the red line reports critical frequencies for amplitudes with monotone-prefix comparability.

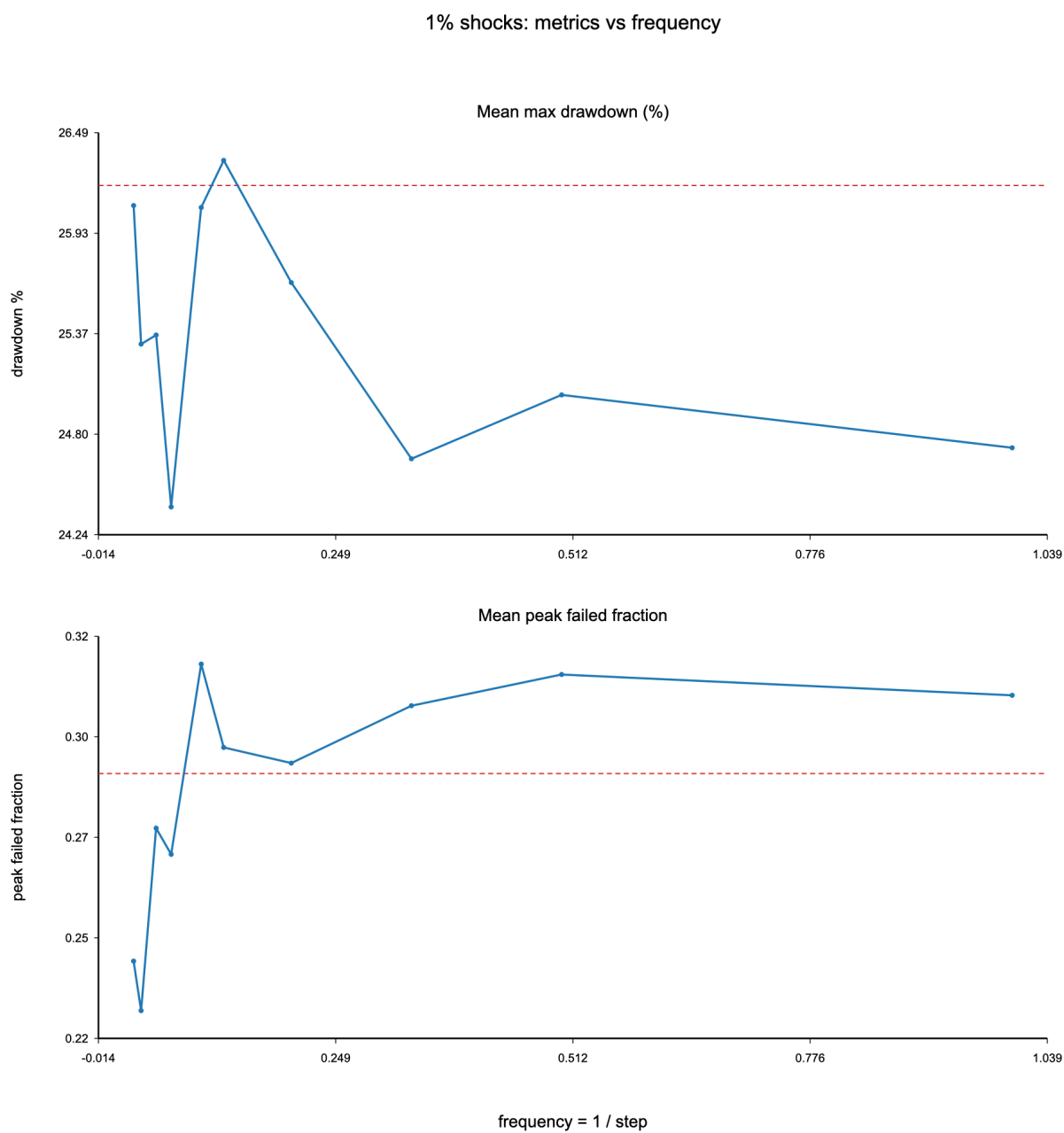


Figure 7.19: Detailed 1% repeated-shock analysis by inter-shock step. Dashed horizontal lines show benchmark levels for drawdown and peak failed fraction.

8 Statistical Validation and Hypothesis Evaluation

This section reports the formal validation of the research hypotheses using run-level outcomes generated by the final repository code. The local-cash-shock validation uses 100 fixed seeds, while the repeated relative-shock validation uses 30 fixed seeds. Bootstrap confidence intervals are computed with 5000 resamples and a fixed bootstrap seed. Pairwise comparisons use bootstrap confidence intervals and nonparametric or paired tests where applicable; multiple comparisons are adjusted using the Holm–Bonferroni correction.

The figures and statistical tests play different roles. Figures illustrate mechanisms and help diagnose model behavior, while the statistical validation checks whether the observed effects are stable across repeated simulation runs. Bootstrap confidence intervals in this section are unadjusted descriptive intervals for mean differences; hypothesis decisions are based on Holm-corrected p -values and on the pre-specified benchmark-like tolerance rule where explicitly stated.

8.1 Validation of H1: localized shocks and propagation beyond the initial region

Hypothesis H1 states that localized shocks can generate failures outside the initially affected interval through endogenous propagation and liquidation. The formal test compares the baseline local-cash-shock scenario with a propagation-suppressed control in which the local contagion component is disabled by setting the interaction radius and contagion parameter to zero.

The statistical results do not support H1 under the final controlled validation. The propagation-enabled baseline does not produce larger outside-region damage than the propagation-suppressed control. The mean cascade indicator, outside failed fraction, spatial extent, and maximum drawdown are all lower in the baseline than in the control, although the differences are not statistically significant after Holm correction.

Table 8.1: Statistical validation of H1: baseline local shock versus propagation-suppressed control.

Metric	Baseline	Control	Diff.	95% bootstrap CI	Holm p
Cascade indicator	0.180	0.280	−0.100	–	1.000
Outside failed fraction	0.0089	0.0131	−0.0042	[−0.0190; 0.0090]	1.000
Spatial extent L	7.27	10.21	−2.94	[−7.09; 1.07]	1.000
Max drawdown, %	3.145	3.486	−0.341	[−1.136; 0.312]	1.000

Note. For the binary cascade indicator, the table reports the risk difference and the Holm-corrected test p -value. Bootstrap confidence intervals are reported for numerical run-level metrics.

These results mean that the illustrative heatmaps should be interpreted as examples of possible propagation-like patterns in individual runs, not as statistical confirmation that the tested local contagion channel systematically increases outside-region damage. Therefore, H1 is not supported in the final controlled validation. This is an informative result: under the current calibration, price-mediated effects and stochastic market dynamics appear to dominate the local contagion component.

8.2 Validation of H2: repeated shocks and benchmark-level damage

Hypothesis H2 states that repeated small shocks can reproduce or exceed the damage of a single benchmark shock when the interval between shocks is sufficiently short. The benchmark is a single -20% relative price shock. Repeated shocks are compared with this benchmark using maximum drawdown, peak failed fraction, and final failed fraction.

The validation gives partial support to H2. Repeated 1% shocks can reproduce benchmark-level damage under the predeclared tolerance criterion, but the frequency response is non-monotonic. Importantly, a non-significant paired test is not interpreted as proof of equality. Benchmark-like status is defined by the tolerance rule, while statistical dominance is assessed by the Holm-corrected pairwise tests.

Table 8.2: Mean outcomes for the main H2 scenarios.

Scenario	n	Mean max drawdown, %	Mean peak failed fraction
Benchmark -20%	30	26.197	0.2873
Repeated 1%, step 5	30	25.653	0.2900
Repeated 1%, step 10	30	26.074	0.3153
Repeated 1%, step 40	30	26.085	0.2393

The repeated 1% step-5 and step-10 regimes are close to the benchmark in mean maximum drawdown. The step-10 regime also produces a higher mean peak failed fraction than the benchmark. However, the paired comparisons do not show statistically significant differences after Holm correction, so the result should be interpreted as benchmark-level comparability rather than strict dominance.

Table 8.3: Paired comparisons for H2 using identical random seeds.

Comparison	Metric	Mean diff.	95% bootstrap CI	Holm p
Step 5 – benchmark	Max drawdown, %	−0.544	[−2.076; 1.088]	1.000
Step 5 – benchmark	Peak failed fraction	0.0027	[−0.0273; 0.0327]	1.000
Step 10 – benchmark	Max drawdown, %	−0.123	[−1.826; 1.737]	1.000
Step 10 – benchmark	Peak failed fraction	0.0280	[−0.0107; 0.0647]	0.969
Step 40 – benchmark	Max drawdown, %	−0.112	[−2.348; 2.472]	1.000
Step 40 – benchmark	Peak failed fraction	−0.0480	[−0.0947; −0.0013]	0.559
Step 5 – step 40	Peak failed fraction	0.0507	[0.0093; 0.0927]	0.278

Note. The confidence intervals are unadjusted descriptive bootstrap intervals. The reported p -values are Holm-corrected paired comparison p -values and are used for significance decisions. Therefore, a confidence interval excluding zero can coexist with a non-significant Holm-corrected p -value.

The critical-frequency scan further supports the partial interpretation of H2. Some amplitudes display a monotone prefix of benchmark-like regimes, while others exhibit non-monotone islands of comparability. This means that temporal clustering matters, but the relationship between inter-shock frequency and damage is not a simple monotone threshold for every shock

amplitude.

Table 8.4: Critical-frequency classification for repeated relative shocks.

Amplitude	Number of shocks	Critical step	Critical frequency	Pattern
0.5%	40	–	–	Islands: steps 1–15 and 30
1.0%	20	–	–	Islands: steps 1–10 and 20–40
2.0%	10	30	0.0333	Monotone prefix: comparable up to step 30
3.0%	7	40	0.0250	Monotone prefix: comparable up to step 40
5.0%	4	40	0.0250	Monotone prefix: comparable up to step 40

Therefore, H2 is partially supported. Repeated small shocks can reproduce benchmark-level damage in several regimes under the pre-specified tolerance rule. This result is interpreted as practical damage comparability within the tested simulation grid rather than as strict statistical equivalence. The paired tests are used to check whether repeated shocks significantly dominate or fall below the benchmark; since they do not show robust dominance after Holm correction, the conclusion remains deliberately conservative.

8.3 Validation of H3a–H3c: thresholds, contagion strength, and interaction radius

For clarity, H3 is evaluated as three sub-hypotheses. H3a states that higher resistance thresholds reduce cascade severity. H3b states that stronger contagion increases fragility. H3c states that wider interaction neighborhoods increase fragility. The final statistical validation tests H3a directly using the implemented `threshold_mu` sweep. H3b and H3c are treated descriptively because the final controlled experimental pipeline does not implement corresponding inferential sweeps over `contagion_beta` and `interaction_radius`.

The threshold component H3a is statistically supported. Higher thresholds reduce cascade outcomes overall, although the relationship is not strictly monotone at the highest threshold level: the cascade indicator rises from 0.06 at $\mu_\theta = 0.35$ to 0.08 at $\mu_\theta = 0.45$, and spatial extent rises from 3.60 to 4.61. The general direction, however, remains clearly downward from the low-threshold regime.

Table 8.5: Mean outcomes across threshold levels.

μ_θ	Cascade indicator	Cascade size C	Outside failed fraction	Spatial extent L
0.15	0.50	0.0922	0.0422	20.56
0.25	0.18	0.0589	0.0089	7.27
0.35	0.06	0.0508	0.0008	3.60
0.45	0.08	0.0485	0.0012	4.61

The key pairwise comparison between the low-threshold regime $\mu_\theta = 0.15$ and the high-threshold regime $\mu_\theta = 0.45$ confirms the expected direction. Higher thresholds reduce cascade size, outside-region failures, and spatial extent. Maximum drawdown, however, does not change significantly.

Table 8.6: Pairwise threshold comparison for H3a: $\mu_\theta = 0.15$ versus $\mu_\theta = 0.45$.

Metric	Difference, low–high	95% bootstrap CI	Holm p
Cascade indicator risk difference	0.420	–	5.95×10^{-10}
Cascade size C	0.0437	[0.0255; 0.0651]	3.82×10^{-4}
Outside failed fraction	0.0410	[0.0228; 0.0622]	8.12×10^{-4}
Spatial extent L	15.95	[11.37; 20.35]	7.34×10^{-10}
Max drawdown, %	0.422	[−0.667; 1.751]	0.500

Note. For the binary cascade indicator, the table reports the risk difference and the Holm-corrected test p -value. Bootstrap confidence intervals are reported for numerical run-level metrics.

Global tests across all threshold levels lead to the same conclusion. Cascade size, outside failed fraction, spatial extent, and the binary cascade indicator depend significantly on μ_θ , while maximum drawdown does not.

Table 8.7: Global tests across threshold levels.

Metric	Welch ANOVA p	Kruskal / chi-square p
Cascade size C	4.70×10^{-5}	5.18×10^{-19}
Outside failed fraction	2.56×10^{-4}	8.14×10^{-18}
Spatial extent L	1.02×10^{-11}	2.64×10^{-17}
Max drawdown, %	0.162	0.229
Cascade indicator	–	1.94×10^{-16}

Therefore, H3a is supported. H3b and H3c are not inferentially validated in the final statistical pipeline. The corresponding radius and beta figures are reproducible and useful as diagnostics, but they should not be interpreted as formal statistical confirmation.

8.4 Final statistical decision

Table 8.8 summarizes the final hypothesis decisions. The strongest statistically confirmed result concerns the role of agent-level thresholds. The repeated-shock frequency mechanism is partially supported through benchmark-level comparability, while the localized-propagation hypothesis is not supported in the controlled ablation test.

Overall, the statistical validation changes the interpretation of the project in an important way. The results do not confirm that the implemented local contagion channel systematically increases outside-region cascade propagation in the baseline local-shock experiment. However, the results do support two central mechanisms: repeated shocks can reproduce benchmark-level market damage in several regimes, and higher agent-level resistance thresholds significantly reduce cascade severity. Thus, the final evidence suggests that systemic fragility in the model is driven more robustly by shock timing and agent-level resilience than by the tested local-propagation ablation.

Table 8.8: Final statistical decision on the research hypotheses.

Hypothesis	Main statistical evidence	Decision
H1	The propagation-enabled baseline does not produce larger outside-region damage than the propagation-suppressed control. The differences in cascade indicator, outside failed fraction, spatial extent, and drawdown are not significant.	Not supported.
H2	Repeated relative shocks can reproduce benchmark-level damage under the tolerance rule in several regimes, but the frequency map is partly non-monotonic and the main paired comparisons are not significant after Holm correction.	Partially supported.
H3a	Increasing μ_θ significantly reduces cascade probability, cascade size, outside failed fraction, and spatial extent overall.	Supported.
H3b	The descriptive β sweep is reproducible but non-monotonic and is not tested inferentially in the final pipeline.	Descriptive only.
H3c	The descriptive radius sweep is reproducible but non-monotonic and is not tested inferentially in the final pipeline.	Descriptive only.

9 Critical Evaluation and Further Research

The model is useful as a controlled experimental environment, but several limitations must be stated explicitly.

First, the one-dimensional lattice is a strong simplification. Real financial exposure networks are multi-layered, directed, weighted, and time-varying. A one-dimensional model helps identify propagation mechanisms, but it cannot fully reproduce the topology of interbank exposures, common portfolios, or fragmented liquidity venues.

Second, the model is not calibrated to empirical market data. Thresholds, shock amplitudes, and contagion parameters are chosen for experimental identification rather than estimated from real balance sheets or trading records. Therefore, the numerical values should not be interpreted as forecasts for a real exchange. The appropriate interpretation is comparative: the experiments identify how outcomes change when shock timing, thresholds, or local interaction parameters are varied.

Third, the controlled validation shows that some visually plausible mechanisms are not statistically supported under the final calibration. In particular, H1 is not supported by the propagation-suppressed ablation test. This does not invalidate the simulator; rather, it identifies a model-design issue for future work. The tested local contagion channel may be too weak, too strongly dominated by price-mediated effects, or insufficiently separated from stochastic market dynamics in the current baseline.

Fourth, H2 is tested using a pre-specified benchmark-like tolerance rule combined with paired seed-level comparisons. This design is appropriate for the objective of the study because the hypothesis concerns practical damage comparability rather than exact statistical equivalence.

A repeated-shock regime is classified as benchmark-like only when its drawdown or peak failed fraction falls within the predefined tolerance margin relative to the single benchmark shock. The paired tests then indicate whether the repeated regime significantly dominates or falls below the benchmark. The resulting interpretation is deliberately conservative: H2 is supported only in the sense of benchmark-level damage comparability within the tested simulation grid.

Fifth, the contagion-strength and interaction-radius components of H3 are treated descriptively rather than inferentially. The visual sensitivity plots are reproducible but non-monotonic, and the final controlled statistical pipeline tests only the threshold component. A stronger validation would include full one-factor sweeps over β and R using the same run-level statistical framework.

Sixth, the current failure rule is absorbing: once an agent fails, it does not recover or re-enter the market. This assumption is defensible for a first cascade model, but it can overstate long-run damage if interpreted as a real market with new entrants, capital injections, or regulatory intervention.

Seventh, the order-book and liquidity layer is still stylized. The fixed-volume spread is a useful improvement because it measures executable liquidity beyond the best quotes, but further work should include cancellation behavior, endogenous market-maker inventory limits, and liquidity replenishment after forced sales.

Future work should proceed in five directions. The first is to refine the local propagation mechanism and retest H1 under calibrations where the direct price channel and the local contagion channel can be separated more cleanly. The second is to implement formal equivalence or non-inferiority tests for H2 and report adjusted confidence intervals for the benchmark-like comparisons. The third is to extend the statistical pipeline to controlled sweeps over β and R , including confidence intervals and effect sizes. The fourth is to replace the one-dimensional lattice with random, small-world, or empirical network topologies. The fifth is to add recovery mechanisms, such as agent re-entry, liquidity-provider intervention, or circuit breakers, and compare whether they reduce cascade probability or only delay failure.

10 Conclusion

This report develops and evaluates a one-dimensional agent-based framework for cascading shocks in an order-driven financial market. The model adds repeated shock events, local cash shocks, stress-based failure, effective thresholds, fixed-volume liquidity measurement, and forced liquidation to a heterogeneous trader simulator. This makes it possible to study not only price responses but also agent-level failure and spatial patterns of stress.

The empirical and statistical results show that shock timing matters. Repeated relative shocks can reproduce benchmark-level market damage in several regimes under a pre-specified tolerance rule, even when each individual shock is small. This result is a tolerance-based benchmark-like classification, not a formal equivalence proof. However, the final validation also shows that the relationship between inter-shock frequency and damage is not universally monotonic: for 0.5% and 1% repeated shocks, benchmark-like outcomes appear in separated islands rather than along

a single critical-frequency boundary. Thus, H2 is partially supported rather than fully confirmed.

Agent-level resilience is the most robust statistically confirmed mechanism in the project. Increasing the mean resistance threshold generally reduces cascade probability, cascade size, outside-region failures, and spatial extent. The relationship is not perfectly monotone at every threshold level, but the overall direction and the formal tests support the threshold-resilience mechanism. Therefore, H3a is supported, while the contagion-strength and interaction-radius components, H3b and H3c, remain descriptive.

At the same time, the localized-propagation hypothesis is not supported by the final controlled ablation test. Although illustrative heatmaps show propagation-like patterns in individual simulations, the propagation-enabled local-shock baseline does not produce statistically larger outside-region damage than the propagation-suppressed control. The final conclusion is therefore nuanced: in the implemented model, systemic risk is shaped most robustly by shock timing and agent-level thresholds, while the tested local contagion channel requires further refinement before it can be claimed as a statistically validated driver of cascades.

For risk analysis, the implication is that repeated mild disturbances should not be treated as harmless simply because each individual shock is small. At the same time, the model demonstrates the importance of formal validation: visually plausible cascade patterns may fail to survive controlled statistical comparison, and this distinction is essential for credible agent-based research.

11 References

- [1] Granovetter, M. (1978). Threshold models of collective behavior. *American Journal of Sociology*, 83(6), 1420–1443. <https://doi.org/10.1086/226707>.
- [2] Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026. <https://doi.org/10.1086/261849>.
- [3] Lux, T., & Marchesi, M. (1999). Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature*, 397(6719), 498–500. <https://doi.org/10.1038/17290>.
- [4] Cont, R., & Bouchaud, J.-P. (2000). Herd behavior and aggregate fluctuations in financial markets. *Macroeconomic Dynamics*, 4(2), 170–196. <https://doi.org/10.1017/S1365100500015029>.
- [5] Watts, D. J. (2002). A simple model of global cascades on random networks. *Proceedings of the National Academy of Sciences*, 99(9), 5766–5771. <https://doi.org/10.1073/pnas.082090499>.
- [6] Farmer, J. D., & Joshi, S. (2002). The price dynamics of common trading strategies. *Journal of Economic Behavior & Organization*, 49(2), 149–171. [https://doi.org/10.1016/S0167-2681\(02\)00065-3](https://doi.org/10.1016/S0167-2681(02)00065-3).
- [7] Chiarella, C., Iori, G., & Perelló, J. (2009). The impact of heterogeneous trading rules on the limit order book and order flows. *Journal of Economic Dynamics and Control*, 33(3), 525–537. <https://doi.org/10.1016/j.jedc.2008.08.001>.
- [8] Gai, P., & Kapadia, S. (2010). Contagion in financial networks. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 466(2120), 2401–2423. <https://doi.org/10.1098/rspa.2009.0410>.
- [9] Battiston, S., Puliga, M., Kaushik, R., Tasca, P., & Caldarelli, G. (2012). DebtRank: Too central to fail? Financial networks, the FED and systemic risk. *Scientific Reports*, 2, Article 541. <https://doi.org/10.1038/srep00541>.
- [10] Cipriani, M., & Guarino, A. (2008). Herd behavior and contagion in financial markets. *The B.E. Journal of Theoretical Economics*, 8(1). <https://doi.org/10.2202/1935-1704.1390>.
- [11] Karimi, F., & Holme, P. (2013). Threshold model of cascades in empirical temporal networks. *Physica A: Statistical Mechanics and its Applications*, 392(16), 3476–3483. <https://doi.org/10.1016/j.physa.2013.03.050>.
- [12] Aymanns, C., & Farmer, J. D. (2015). The dynamics of the leverage cycle. *Journal of Economic Dynamics and Control*, 50, 155–179. <https://doi.org/10.1016/j.jedc.2014.09.015>.

- [13] Acemoglu, D., Ozdaglar, A., & Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2), 564–608. <https://doi.org/10.1257/aer.20130456>.
- [14] Glasserman, P., & Young, H. P. (2016). Contagion in financial networks. *Journal of Economic Literature*, 54(3), 779–831. <https://doi.org/10.1257/jel.20151228>.
- [15] Bookstaber, R., Paddrik, M., & Tivnan, B. F. (2018). An agent-based model for financial vulnerability. *Journal of Economic Interaction and Coordination*, 13(2), 433–466. <https://doi.org/10.1007/s11403-017-0188-1>.
- [16] Barucca, P., Bardoscia, M., Caccioli, F., D’Errico, M., Visentin, G., Caldarelli, G., & Battiston, S. (2020). Network valuation in financial systems. *Mathematical Finance*, 30(4), 1181–1204. <https://doi.org/10.1111/mafi.12272>.

Appendix A: Disclosure of AI assistance

The following generative-AI tools were used during the preparation of this report: ChatGPT and OpenAI Codex. The table below records representative prompts and summarizes the type of assistance received. The research topic, literature review, model design, simulation results, statistical outputs, and final interpretation remain the author’s own responsibility.

Table A.1: Representative prompts and AI assistance.

# / Tool	Representative prompt	Summary of response and use
1. ChatGPT	“Explain the theoretical connection between threshold cascade models, localized shocks, and endogenous amplification in agent-based financial markets. Focus only on the conceptual mechanism.”	The model clarified how threshold failures, local interactions, and liquidation feedback can generate cascade-like dynamics. This was used only to refine the theoretical framing. The literature review, source selection, and discussion of previous studies were compiled by the author independently.
2. OpenAI Codex	“Review the plotting code and suggest formatting improvements for the report figures. Focus only on readability: figure size, axis labels, captions, margins, colors, and legends. Do not change the simulation logic or data.”	Codex helped identify figure-formatting issues, such as excessive white space, small axis labels, and unclear captions. The assistance was limited to visual presentation. The simulation outputs and numerical results were not altered.
3. ChatGPT	“I have already selected candidate statistical tests: bootstrap confidence intervals, paired comparisons, Fisher or chi-square tests for binary outcomes, and Holm–Bonferroni correction. Please check whether these tests are methodologically appropriate for my run-level simulation metrics. Do not run tests or generate numerical results.”	The model commented on the suitability and interpretation of the selected tests, especially for benchmark-like comparisons and non-significant results. All statistical calculations, code execution, numerical results, and final interpretation were performed by the author independently.